

Chapter One

Introduction to Operations Research

- The subject OPERATIONS RESEARCH is a branch of applied mathematics.
- OR used to provide a scientific base for management.
- Operations research is the body of knowledge:
 - Which uses mathematical techniques to solve management problems and make timely optimal decisions.
- Operations research is concerned with helping managers and executives to make better decisions.
- OR avoid the dangers from taking decisions merely by guessing or by using rules of thumb .

1.1. The History of Operations Research

- No science has ever been born on a specific day
- Operations Research is a ‘war baby’.
- Because operations research (O.R.) was coined during World War II,
- When the British military management called upon a group of scientists together to apply a scientific approach in the study of military operations to win the battle.
- The main objective was to allocate scarce resources in an effective manner to various military operations.
- It is now widely used in military, business, industry, transportation, public health, crime investigation, etc. for quality decision making purpose.

1.2. Definition, Application and Objectives of Operations Research

- According to **Morse and Kimball**:
 - ✓ Operations research is a scientific method of providing for executives departments with a quantitative basis for decisions regarding operations under their control.
- According to **Miller and starr**:
 - ✓ Operations research is defined as the application of decision theory. It uses a scientific, mathematical, or logical means to cope with the problems.
- According to H.M Wagner:
 - ✓ OR Is a scientific approach to problem solving for executive management.
- According to the operations Research of America:
 - ✓ Operations Research is an experimental and applied science devoted to observing, understanding and predicting to behavior of purposeful man machine systems.

The common ideas emerged out of all such definitions are:

- The use of the scientific method,
- Study of the complex relationships, and
- Provision of a basis for decision making.

1.3. The Nature and Significance of Operations Research

- The application of operations Research is different for the different fields.
- Operations Research is useful in every field of human activities, where optimize of resources is required.

Areas of Application

- National planning and Budgeting
- Defense services Operations
- Engineering Activities
- Business Management and Completion
- Agriculture and Irrigation
- Education and Training
- Transportation and Communication

1.4. The Scope and Features of operations Research

Scope of operations Research

The major scope of operations research are:

- Cash Management
- Inventory Control
- Aid to decision making and improve its quality,
- Integrating the system as a whole,
- Improve objectivity of analysis and clarity of thought.
- Minimize the cost and maximize the profit.
- Improve proclivity and efficiency, and success in competitions and market leadership.

Features of operations Research

- Provision of scientific Basis to Decision Making
- Conversion of complex problems into simple problems
- **Work for Trade~Offs**
- **Provision of system integration**
- Optimization of Researches
- **Minimization of cost**
- **Maximization of profit**
- **Selection of Best Alternatives**

1.5. Operations Research Approach to Problem Solving and Model Building.

Operation Research Approach to Problem Solving
Contain the following seven fundamental stages.

- Understand and formulate the problem
- Determine assumptions
- Formulate the problem in a mathematical frame work
- Solve the model of the problem
- Interpret the Results of the model
- Validate the model
- Implement the model ~ predict and decide

Model Building.

What is model?

- ✓ Model is an idealized representation of the real life situation.
- ✓ A model is a schematic description of a system, theory, or phenomenon that accounts for its known or inferred properties and may be used for further study of its characteristics.
- ✓ It represents one or a few aspects of reality.
- ✓ The objective of the model is to provide a means for analysing the behaviour of the system for purpose of improving its performance

Significance and advantage of using models

- Some of the major advantages to be gained from using models are:
- They assist in the understanding of the problem;
- They give a clear picture of the problem;
- They assist to modify the existing design or system;
- They specify inputs, outputs, objectives and constraints of the problem;
- Models can be more easily manipulated and analyzed the real system;
- They provide an indication of the information flow and gaps therein; and
- They are fundamental tools in decision making a tool to help managers to decide on what questions to ask themselves.

Characteristics and Types of Models

Characteristics of Models

- Faithful production or representation of real situation
- Flexible enough to incorporate modification and changes
- Simple and easy to understand
- Capable of manipulation by changing, inputs and other parameters and getting corresponding outputs.
- Minimum number of assumptions and variables, and
- Clear definition of parameters including assumptions

Types of Models

Based on various schemes models can be classified as follows:

1. **By Degree of abstraction.**
 - a. **Mathematical Models :** Linear programming problem and transportation problem are most abstract since they require great concentration to get the idea of the real life situation in addition to the mathematical knowledge.
 - b. **Language Models:** Concrete models (model of earth, dam, building) are the least abstract since they instantaneously suggest the shape of the modeled entity.

2. By Function

a. Descriptive Models:

- Explains operations in non mathematical language and try to define the functional relationship between various operations.
- Do not predict system's behavior. Example organizational chart, pie diagram and layout plan.

b. Predictive Models: Predict the behavior of the system.

- Example Exponential smoothing forecast model predicts the future demand.

c. Normative or Prescriptive Models:

- Develops decision rules or criteria for optimal solutions. Example Linear Programming.

3. By structure

a. **Physical [Iconic] models:** Physical representations of the phenomena under analysis.

- Properties of the real system represented by the properties themselves, frequently with change of scale. Example Globes, Solar system etc.

b. Analogue models:

- They use one set of properties to represent some other set of properties which the system under study possesses.
- Revealing inter and intra relationships between two or more parameters.
- Represent dynamic situations

Example: Graphs is a well suitable for representing quantitative relationship between any two properties and predicts how a change in one property affects the other.

c. Symbolic [mathematical] models: use of mathematical and statistical notations to represent the problem under analysis.

4. By Nature of the environment

a. Deterministic Models

- Variables are completely defined and the out comes are **certain**.
- Economic Order Quantity(E.O.Q), Linear Programming, transportation and assignment problems.

b. Probabilistic Models

- They are the product of an environment of risk and **Uncertainty**.
- Take the form of probability distribution.
- Example, Exponential smoothing forecast model, game theory.

5. By Extent of generality

- a. **General Models:** A model that are used for a number of functions or generally applicable model.
 - Example Linear Programming model.
- b. **Specific models :** Are models that are applicable only in specific function.
 - Example sales response curve or equation as a function of advertising is applicable only in marketing function.

6. By the time of horizon

- a. **Static Models:** They are one time decision model. Example E.O.Q
- b. **Dynamic Models:** Time plays an important role
 - Used for optimization multistage decision problems
 - Example Dynamic programming.

1.7. Methodology and Techniques in Operations Research

1. Analytical Method [Deterministic Method]

2. Trial and Error Method

3. Simulation Method [Stochastic Method]

1.9. Appropriateness and Limitations of Operations Research

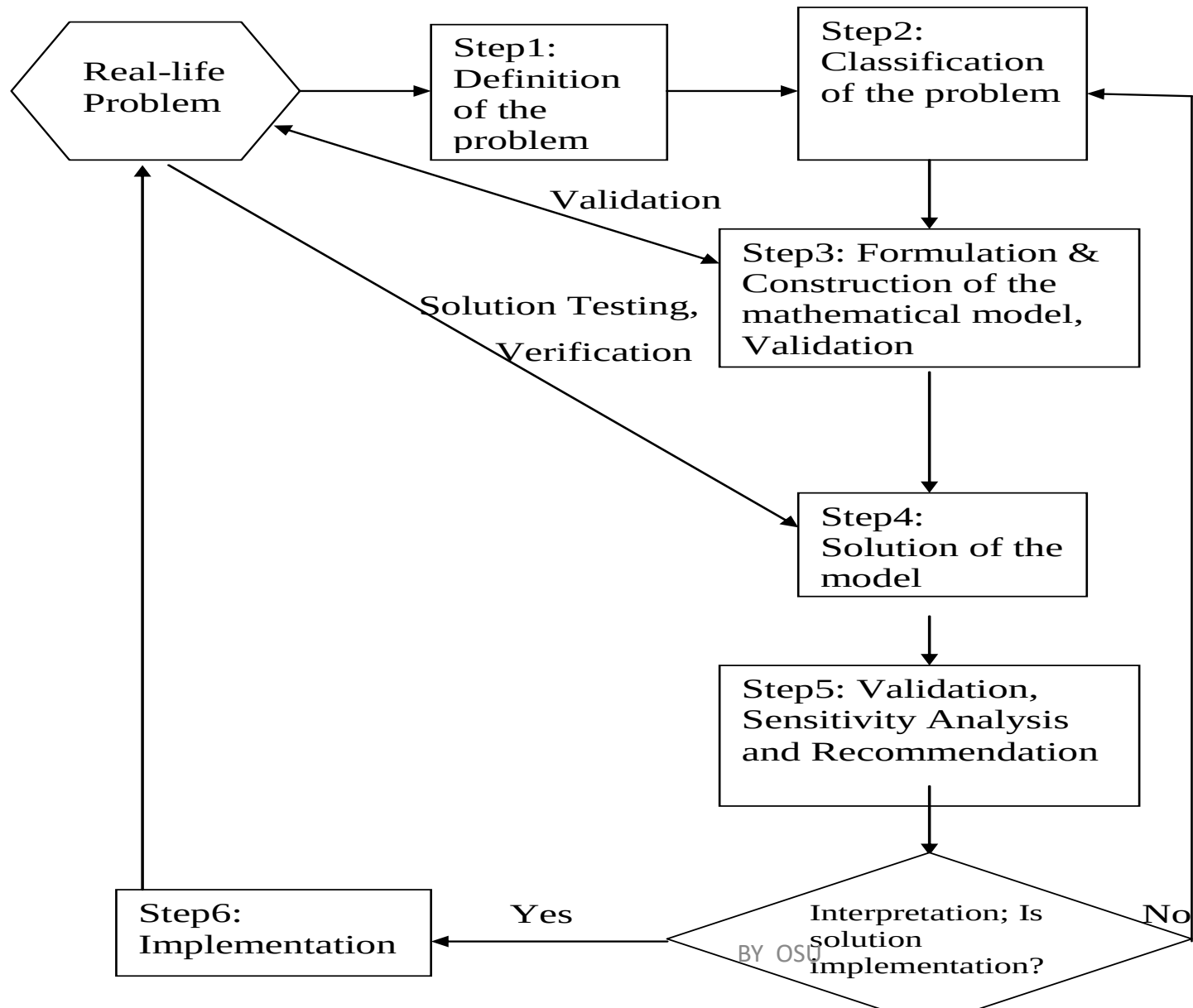
Appropriateness of Operations Research

- Production Management
- Materials Management
- Quality Management
- Service Industry
- Maintenance Management
- Project Management

Limitations of Operations Research

- Information Gap
- Quantification Techniques
- Finite Variables
- Limited Number of Constraints

Process of Operations Research



Chapter Two

Linear Programming Problem

2.1. Introduction

What is Linear Programming(LP)?

- LP is a tool for solving optimization problem.
- LP is model, deal with allocation problems: to determine an optimal allocation of an organization's limited resources.
- LP Is used for solving problems arising in business, government, industry, hospitals, etc.
- LP is one of the most powerful and useful techniques for making managerial decisions.

Assumptions

Certainty, linearity, fixed technology, constant profit per unit, Nonnegative and Static Model.

- **Certainty:** Refers to the objective function coefficient, the right hand side (R.H.S) or quantity, coefficients of the constraint, and resources in the constraint are precisely known and that their values do not change with time.
- **Linearity :** Higher power of the variables do not appear in the expression for the objective function as well as constraints.
- They do not have expression like x_1^2 , x_2^3 , x_1x_2 , $a_1x_1+a_2\log x_2$.
- Linearity is expressed as $a_1x_1 + a_2x_2 + a_3x_3 + \dots + a_n x_n$

- The term **linearity** implies **proportionality** and **additivity**.
- **Proportionality** refers to if a product yields a profit of Birr 5, the profit earned from the sale of 10 such product will be ($5 \times 10 = \text{Birr } 50$). However, the real world situation may not strictly linear.
- **Additivity** : Means that if we use t_1 hours on machine A to produce product I and t_2 to make product II, the total time required to make product I and II on machine A is $t_1 + t_2$ hours.
- However, some process may not behave in this way. Example, when several liquids of chemical compositions are mixed, the resulting volume may not be equal to the sum the volumes the individual liquids.

- **Fixed technology:** Refers to the fact that the production requirements are fixed during the planning period and will not change in the period
- **Constant profit per unit** It is assumed that the *profit contribution of a product remains constant*, irrespective of level of production and sales.
- **Nonnegative** : All variables are restricted to nonnegative values (i.e., their numerical value will be ≥ 0).
- **Static Model:** It is assumed that only one decision is required for the planning period.
- This implies that the linear programming problem is a single stage decision problem.

Terminology

a. Optimization

- LP attempts to either maximize or minimize the value of the objective ~*function.

b. Profit or cost coefficients

- They are the coefficients of the variables in the objective function.

c. Constraints

- Expressed as linear inequalities and/or equations.
- Reflect the fact that resources are limited in the blending problem.

d. Input-output coefficients

- They are coefficients of the constraints' variables.

e. Capacities

- Expressed as some upper or lower limit.
- On the right-hand side of the constraints.

g. Programme: *is referred as a course of action covering a specified period of time, say planning period.*

h. Optimal course of action or optimal solution: A programme is optimal, when it *maximizes or minimizes* some measure or criterion of effectiveness, such as profit, sales or costs.

i. Programming: *refers to a systematic procedure by which a particular program or plan of action is designed.*

j. Candidates or activity: refers to number of products or any such items, which need the utilization of available resources in a certain required proportion.

k. Available resources: any nature, such as money, area of land, machine hours, and man-hours or materials.

2.2. Structure of Linear programming problem

(General Linear Programming Problem)

- A general mathematical way of representing a Linear Programming Problem (L.P.P.) is as given below:

$$\text{Max or Min } z = c_1x_1 + c_2x_2 + c_3x_3 + \dots + c_nx_n \text{ (Objective Function)}$$

$$\text{Subject to : } a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \leq \text{or } = \text{or } \geq b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \leq \text{or } = \text{or } \geq b_2$$

$$\dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \leq \text{or } = \text{or } \geq b_m$$

$$\text{all } x_i \geq 0, \quad i = 1, 2, 3, \dots, n$$

(non-negativity constraints).

- Generally $\leq$ sign is used for maximization problems and $\geq$ sign is used for minimization problems and in some problems, which are known as mixed problems we may have all the three signs.

The steps for formulating the linear programming are:

1. Identify the unknown decision variables to be determined and assign symbols to them.
2. Identify all the restrictions or constraints in the problem and express them as linear equations or inequalities of decision variables.
3. Identify the objective or aim and represent it also as a linear function of decision variables.

Properties of Linear Programming Model

- a) The model must have an objective function.
- b) The model must have structural constraints.
- c) The model must have non-negativity constraint.

2.3. Application Area of linear Programming Problem

Linear programming is one of the most widely applied technique of operations research in business, industry, and numerous other fields.

A few areas of its applications are given below:

1. Industrial Application

- a. Product Mix Problem
- b. Blending Problem
- c. Production scheduling problems
- d. Trim Loss problems
- e. Assembly lines problems
- f. Make or Buy (subcontracting problems)

2. Management applications

- a. Media selection problems
- b. Portfolio selection problems
- c. Profit planning g problems
- d. Transportation problems
- e. Assignment problems
- f. Man power scheduling problems

3. Miscellaneous Applications

- a. Diet problems
- b. Agricultural problems
- c. Fight scheduling problems
- d. Environmental protection
- e. Facilities Location problems

2.4. Methods for the Solution of a Linear Programming Problem

The various methods available to solve the problem are:

1. **The Graphical Method:** Also known as geometrical method.
 - Provides a pictorial representation of the solution process.
2. **The Systematic Trial and Error method,** where we go on giving various values to variables until we get optimal solution.
 - This method takes too much of time and laborious, hence this method is not discussed here.

3. The Vector method. In this method each decision variable is considered as a vector and principles of vector algebra is used to get the optimal solution.

This method is also time consuming, hence it is not discussed here.

4. The Simplex method. When the problem is having more than two decision variables, simplex method is the most powerful method to solve the problem.

It has a systematic programme, which can be used to solve the problem.

. One problem with two variables is solved by using both graphical and simplex method, so as to enable the reader to understand the relationship between the two.

2.4.1. Graphical Method

The characteristics of Graphical method are:

- (i)* Generally the method is used to solve the problem, when it involves two decision variables.
- (ii)* For three or more decision variables, the graph deals with planes and requires high imagination to identify the solution area.
- (iii)* Always, the solution to the problem lies in first quadrant.
- (iv)* This method provides a basis for understanding the other methods of solution.

The graphical method consists the following steps:

1. Represent the given problem in the mathematical form,
2. Draw the x and y axes,
3. Plot each the constraint on the graph,
4. Identify the feasible region(solution space) that satisfies all the constraints simultaneously,
5. Find the coordinates of each extreme point using extrema enumeration approach.

Maximization Models

- It has an objective function.
- It has structural constraints.
- The relationship between variables and constraints are linear.
- It has non-negativity constraint
- The coefficients of variables may be positive or negative or zero.
- The inequalities are of $\leq$ type.
- At most term is used

Example~1~

- A company manufactures two products, X and Y by using three machines A, B, and C. Machine A has 4 hours of capacity available during the coming week. Similarly, the available capacity of machines B and C during the coming week is 24 hours and 35 hours respectively. One unit of product X requires one hour of Machine A, 3 hours of machine B and 10 hours of machine C. Similarly one unit of product Y requires 1 hour, 8 hour and 7 hours of machine A, B and C respectively. When one unit of X is sold in the market, it yields a profit of Birr 5/~ per product and that of Y is Birr 7/~ per unit. Solve the problem by using graphical method to find the optimal product mix.

Machines	Products (Time required in hours).		Available capacity in hours.
	X	Y	
A	1	1	4
B	3	8	24
C	10	7	35
Profit per unit in Birr	5	7	

Let the company manufactures x units of X and y units of Y , and then the L.P. model is:

Maximize $Z = 5x + 7y$ s.t.

$$1x + 1y \leq 4$$

$$3x + 8y \leq 24$$

$$10x + 7y \leq 35 \text{ and}$$

Both x and y are ≥ 0 .

As we cannot draw graph for inequalities, let us consider them as equations.

Maximize $Z = 5x + 7y$ s.t.

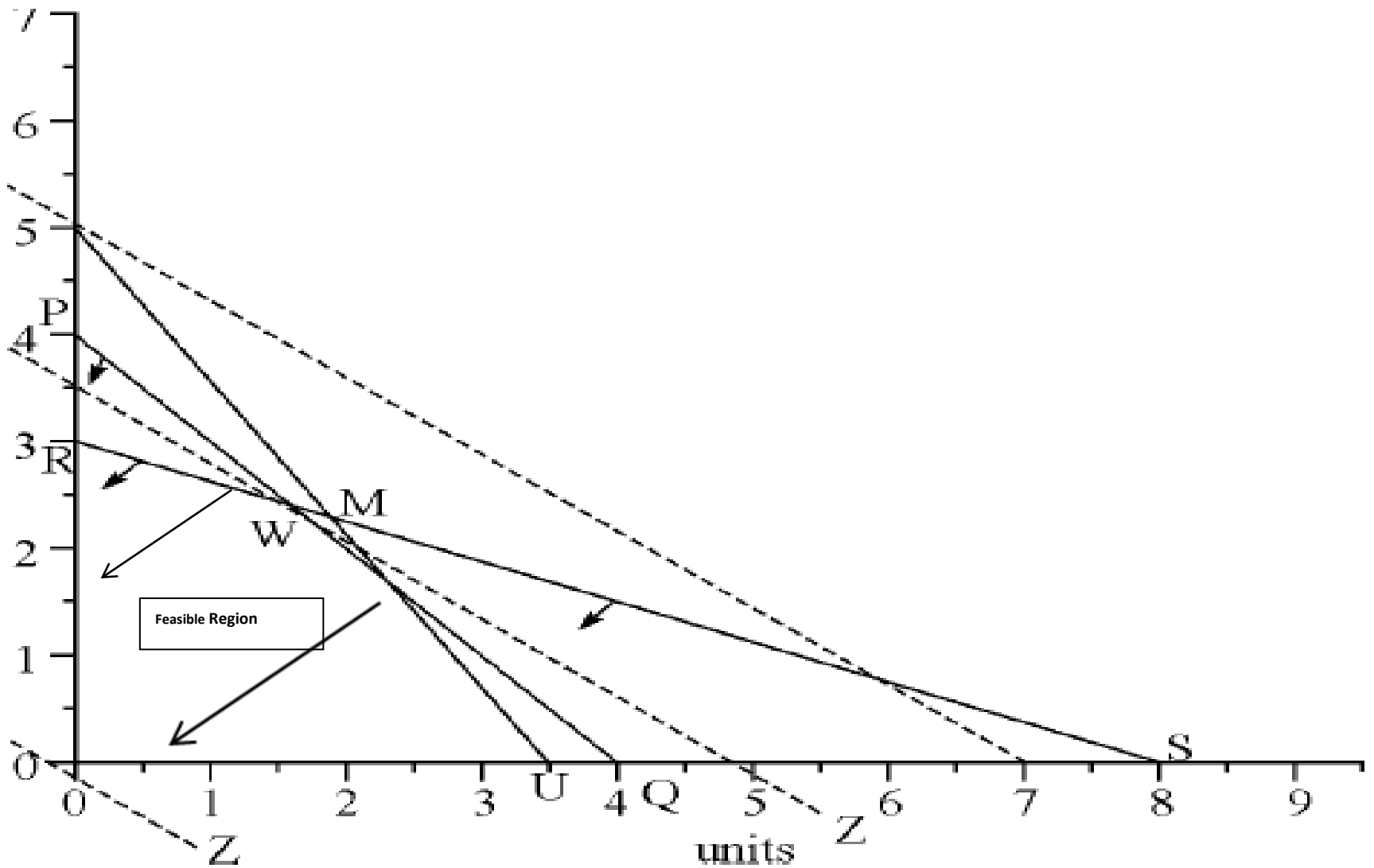
$$1x + 1y = 4$$

$$3x + 8y = 24$$

$$10x + 7y = 35 \text{ and both } x \text{ and } y \text{ are } \geq 0$$

- Let us take machine A. and find the boundary conditions. If $x = 0$, machine A can manufacture
- $4/1 = 4$ units of y .
- Similarly, if $y = 0$, machine A can manufacture $4/1 = 4$ units of x . For other machines:
- Machine B When $x = 0$, $y = 24/8 = 3$ and when $y = 0$ $x = 24/3 = 8$
- Machine C When $x = 0$, $y = 35/10 = 3.5$ and when $y = 0$, $x = 35 / 7 = 5$.
- These values we can plot on a graph, taking product X on x-axis and product Y on y- axis.

The graph is:



- In maximization problem, we select the co-ordinates giving maximum value.
- And in minimization problem, we select the co-ordinates, which give minimum value. In the problem the co-ordinates of the corners are:
- $R = (0, 3.5)$, $O = (0,0)$, $U = (3.5,0)$, $V = (2.5, 1.5)$ and $W = (1.6,2.4)$. Substituting these values in objective function
- $Z(0,3.5) = 5 \times 0 + 7 \times 3.5 = \text{Birr } 24.50$, at point R
- $Z(0,0) = 5 \times 0 + 7 \times 0 = \text{Birr } 00.00$, at point O
- $Z(3.5,0) = 5 \times 3.5 + 7 \times 0 = \text{Birr } 17.5$ at point U
- $Z(2.5, 1.5) = 5 \times 2.5 + 7 \times 1.5 = \text{Birr } 23.00$ at point V
- $Z(1.6, 2.4) = 5 \times 1.6 + 7 \times 2.4 = \text{Birr } 24.80$ at point W

- Hence the optimal solution for the problem is company has to manufacture 1.6 units of product X and 2.4 units of product Y, so that it can earn a maximum profit of Birr 24.80 in the planning period.

Example ~2~

Consider the following linear programming problem and solve using graphical method

Maximize $Z = 10x + 10y$ S.T.

$$1x + 2y \leq 2000$$

$$1x + 1y \leq 1500$$

$$0x + 1y \leq 600 \text{ and both } x \text{ and } y \text{ are } \geq 0.$$

Minimization Model

- It has an objective function.
- It has structural constraints.
- The relationship between variables and constraints are linear.
- It has non-negativity constraint
- The coefficients of variables may be positive or negative or zero
- The inequalities are of $\geq$ type.
- At least term is used

Example 1

- A patient consults a doctor to check up his ill health. Doctor examines him and advises him that he is having deficiency of two vitamins, vitamin A and vitamin D. Doctor advises him to consume vitamin A and D regularly for a period of time so that he can regain his health. Doctor prescribes tonic X and tonic Y, which are having vitamin A, and D in certain proportion. Also advises the patient to consume at least 40 units of vitamin A and 50 units of vitamin Daily. The cost of tonics X and Y and the proportion of vitamin A and D that present in X and Y are given in the table below.

Vitamin	Tonics		Daily requirement in units.
	X	Y	
A	2	4	40
D	3	2	50
Cost in Birr. per unit.	5	3	

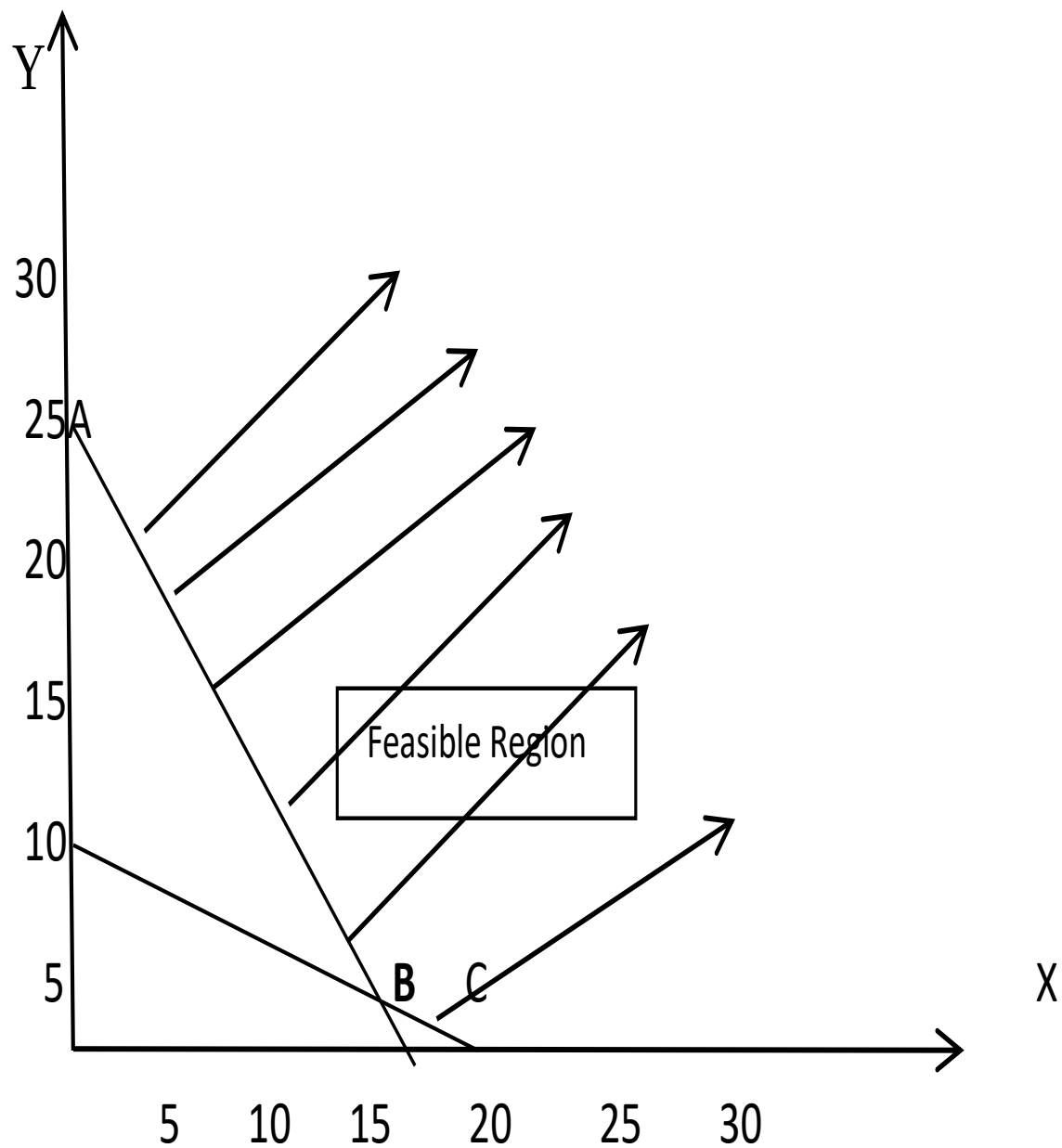
- a. Formulate l.p.p. to minimize the cost of tonics.
 - b. Solve the linear programming problem using graphical method
- Solution: Let patient purchase x units of X and y units of Y .
 - Objective function: Minimize $Z = 5x + 3y$
 - Inequality for vitamin A is $2x + 4y \geq 40$ (Here at least word indicates that the patient can consume more than 40 units but not less than 40 units of vitamin A daily).
 - Similarly the inequality for vitamin D is $3x + 2y \geq 50$.

- For non–negativity constraint the patient cannot consume negative units. Hence both x and y must be ≥ 0 . Now the l.p.p. model for the problem is:
- Linear Programming Models (Resource Allocation Models)
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- Minimize $Z = 5x + 3y$ s.t.
- $2x + 4y \geq 40$
- $3x + 2y \geq 50$ and
- Both x and y are ≥ 0 .
- Solve the linear programming problem using graphical method
- Interpret the result.

- NB. The graph of the minimization problem should be shaded up ward because of the $\geq$ inequalities.

Solution

- As we cannot draw graph for inequalities, let us consider them as equations.
- Minimize $Z = 5x + 3y$ s.t.
- $2x + 4y = 40$
- $3x + 2y = 50$
- Both x and y are ≥ 0
- Let us take Vitamin A. and find the boundary conditions. If $x = 0$, tonic X can be consumed
- $40/4 = 10$ of tonic y .
- Similarly, if $y = 0$, Vitamin A can be consumed $40/2 = 20$ of tonic x .



- Here we find the co-ordinates of corners of ABC and substitute the values in the objective function. In minimization problem, we select the co-ordinates giving minimum value. In the problem the co-ordinates of the corners are:

$A = (0, 25)$, $B = (15, 2.5)$, $C = (20, 0)$, Substituting these values in objective function

- $Z(0, 25) = 5 \times 0 + 3 \times 25 = \text{Birr } 75$ at point A
- $Z(15, 2.5) = 5 \times 15 + 3 \times 2.5 = \text{Birr } 82.5$, at point B
- $Z(20, 0) = 5 \times 20 + 3 \times 0 = \text{Birr } 100$ at point C
- Hence the optimal solution for the problem is the patient has to consume 0 units of tonic X and 25 units of tonic Y, so that he can minimize his cost to Birr 75.

Example 2. Consider the problem:

Minimize $Z = 15x + 25y$ S.T.

$$10x + 30y \geq 300$$

$$10x + 40y \geq 200 \text{ and both } x \text{ and } y \geq 0$$

a. Solve the problem using graphical method

b. Interpret the solution

2.4.2. Special Case in LPP of Graphical Solution

Some special issues that may arise during the solution LP problems in a graphical procedure.

These issues are:

- a. Problems with no feasible solutions;
- b. Unbounded solution;
- c. Redundancy in constraints; and
- d. Problems with multiple optimal solutions

- a. Problems with no feasible solutions
- Occurs in problems that have a mix of greater than constraints, and less than constraints.
 - where in order to satisfy one of the constraints, another constraint must be violated.
 - This situation sometimes happens when the problem has been formulated incorrectly.
 - When such an issue is observed in an LP problem, the manager needs to give time to check one of the following errors. The correctness of the coefficients of the constraints, if there is a plotting error, and if an inequality has the wrong direction.

Minimization Problem.....

Example

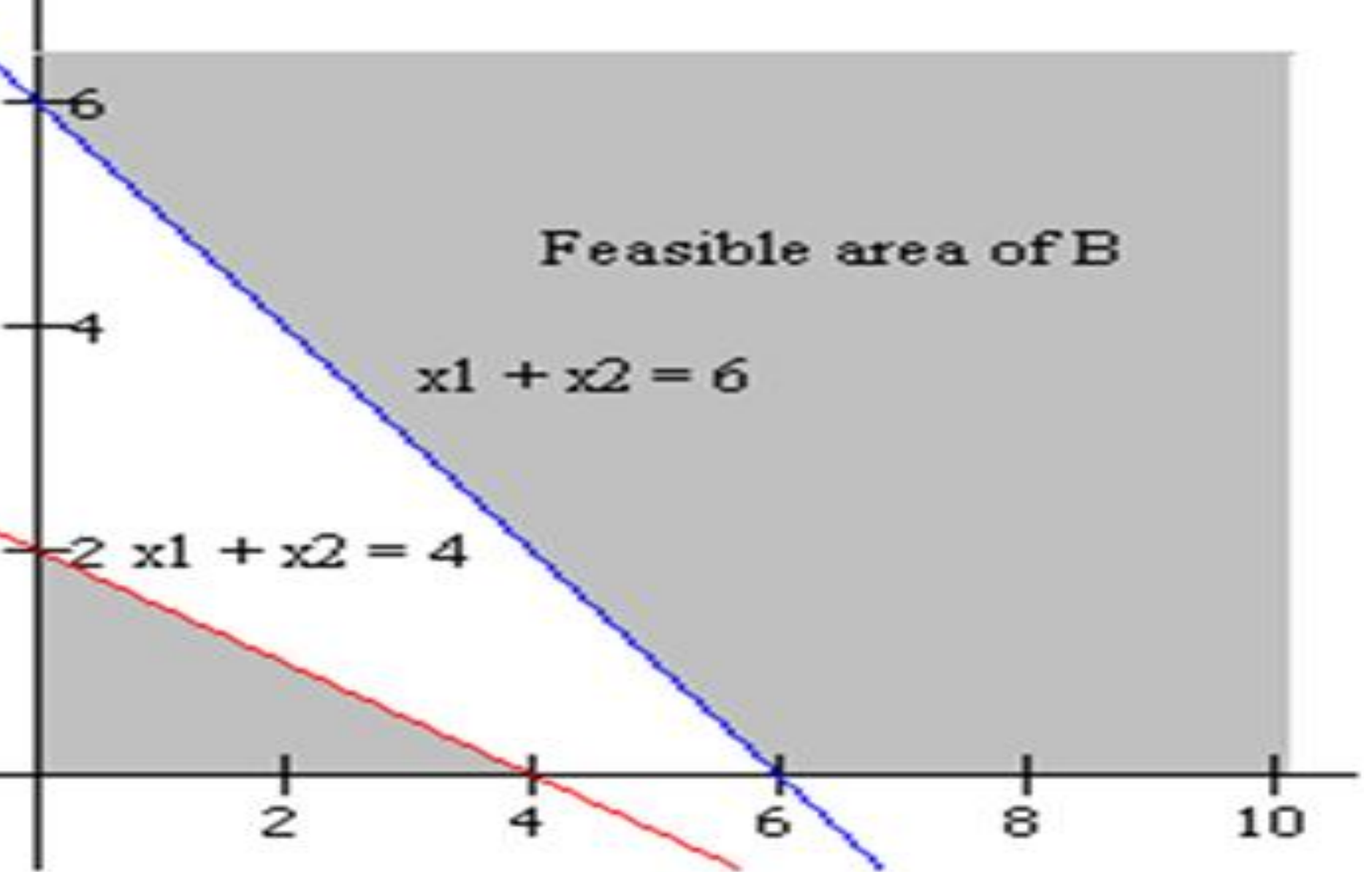
Find the maximum value of $Z = 4x_1 + 3x_2$

Subject to the constraints

$$\text{A: } x_1 + x_2 \geq 6$$

$$\text{B: } x_1 + 2x_2 \leq 4$$

All variables ≥ 0



The graph reveals that no combination of x_1 and x_2 can simultaneously satisfy both constraints, and there is no feasible solution space for the

b. Unbounded Problems

- An unbounded problem exists when the value of the objective function can be increased without limit.
- This difficulty usually arises due to incorrectly maximizing the problem when the real goal is to minimize an objective function or using a relation that constraints when less than constraints are called for.

Example

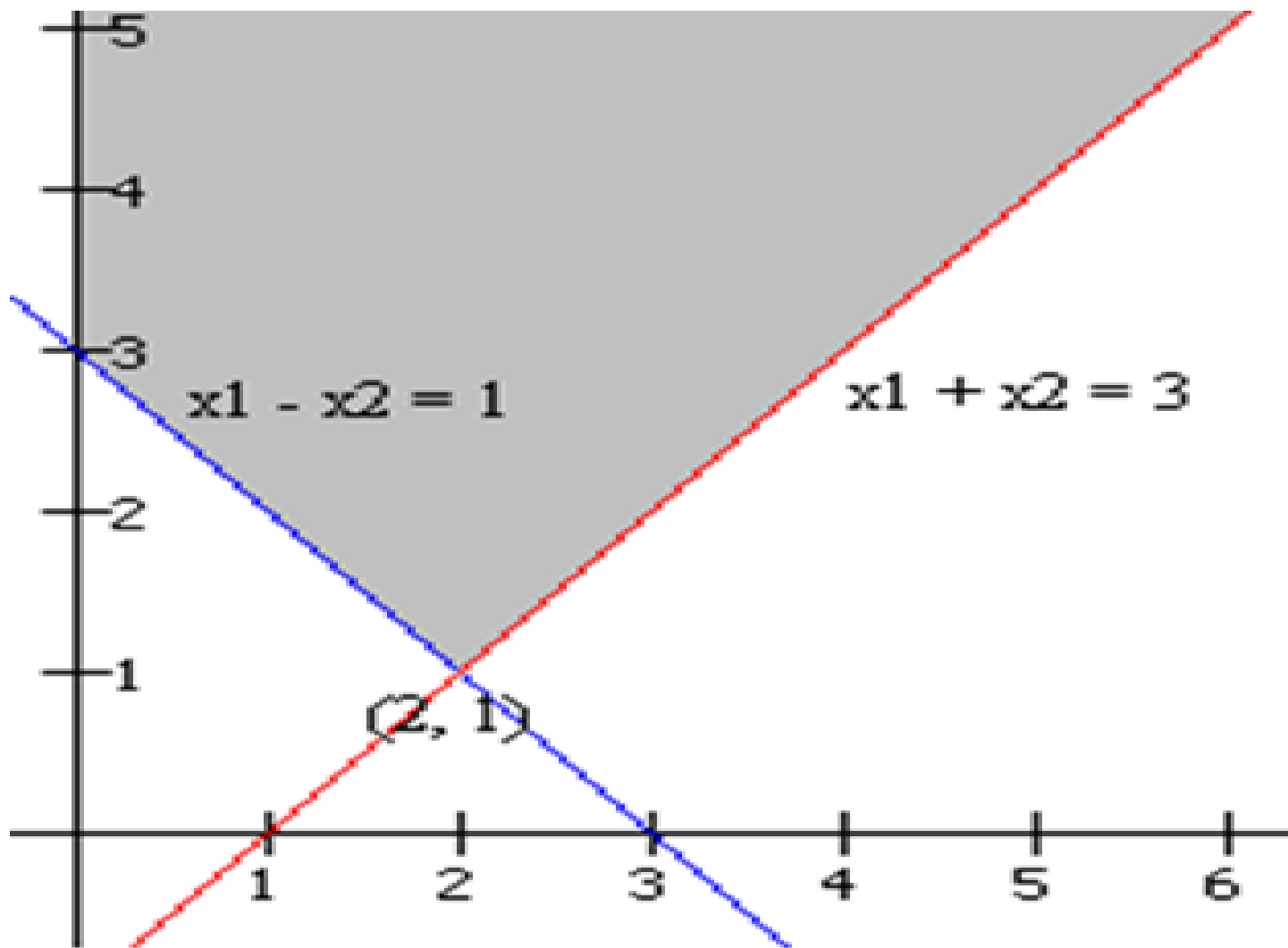
Maximize $Z = 3x_1 + 2x_2$

Subject to the constraints

1. $x_1 - x_2 \leq 1$

2. $x_1 + x_2 \geq 3$

All variables ≥ 0



The graph reveals that the feasible solution space has exactly two corner points with coordinates $(2, 1)$ and $(0, 3)$. The rules discussed under the graphic methodology would be violated because there are many more points other than the above identified corner points where the value of the objective is more. Since the feasible solution space is unbounded, we are able to locate points at infinity. Hence, the problem is said to have no unbounded solution. That is, it has a feasible solution but is unbounded.

c. Redundancy in Constraints

- In some cases a constraint does not form a unique boundary of the feasible solution space.
- Such a constraint is known as a redundant constraint.
- Its removal would not change the feasible solution space.

Maximize $Z = 10x_1 + 12x_2$

Subject to the constraints

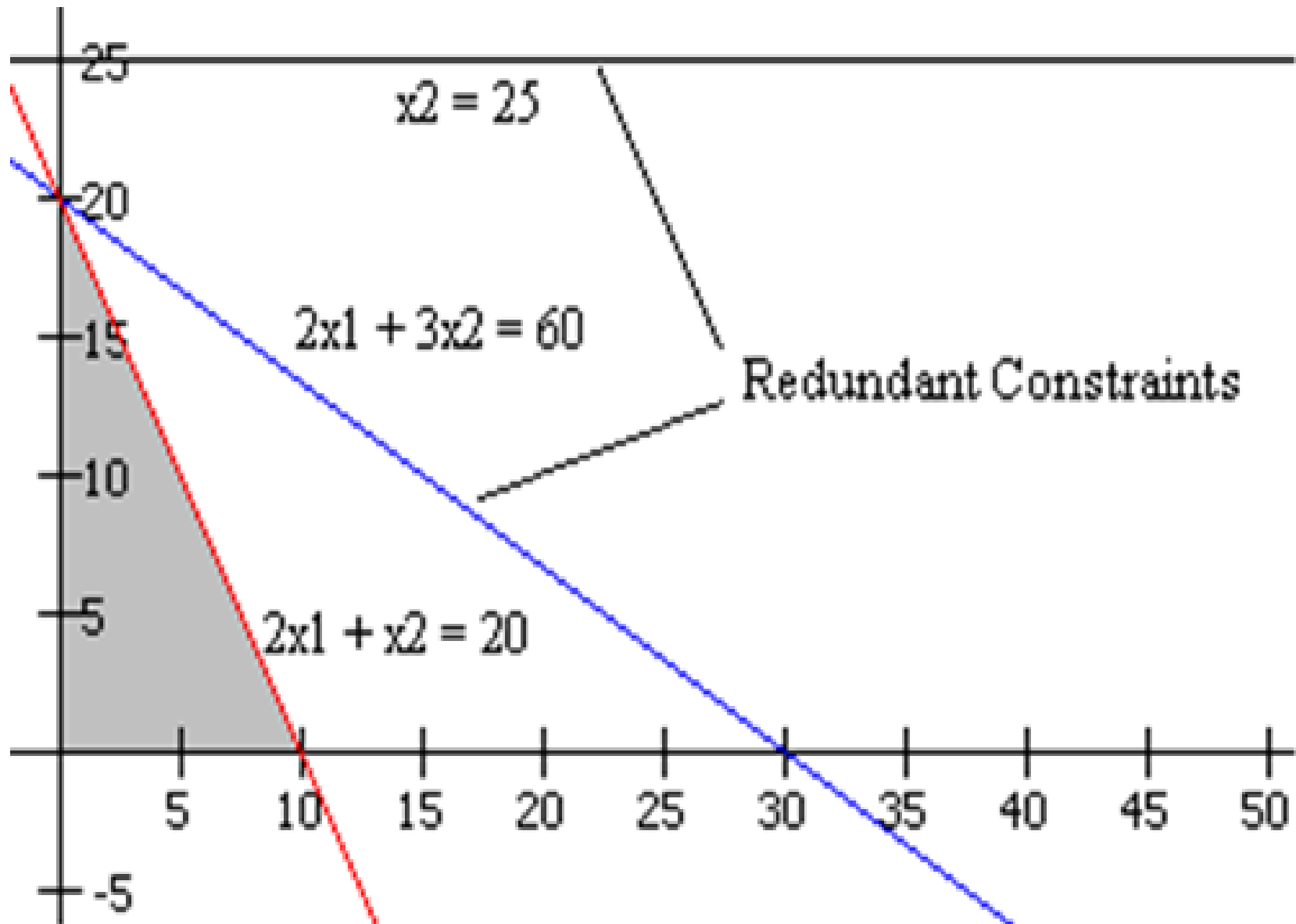
$$2x_1 + 3x_2 \leq 60$$

$$2x_1 + x_2 \leq 20$$

$$x_2 \leq 25$$

All variables ≥ 0

Redundant solution



- Some LP problems, it might be observed multiple optimal solutions.

Maximize $Z = 60x_1 + 30x_2$

Subject to the constraints

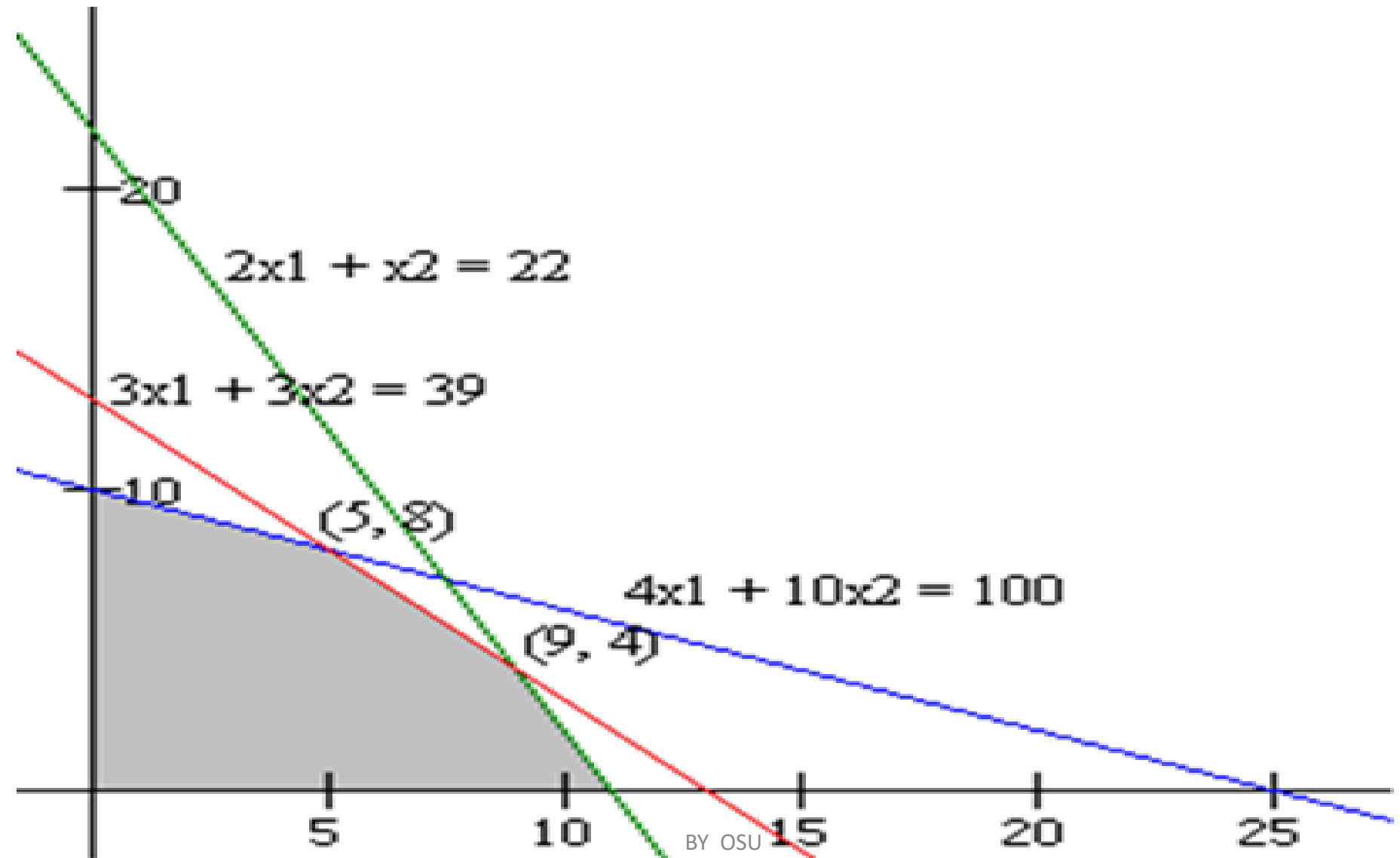
Assembly: $4x_1 + 10x_2 \leq 100$

Inspection: $2x_1 + 1x_2 \leq 22$

Storage: $3x_1 + 3x_2 \leq 39$

All variables ≥ 0

Multiple solution



- It was revealed on the graph that the feasible solution space has exactly five corner points with coordinates (0,0), (11,0), (9,4), (5,8) and (0,10).
- By the rules discussed under the graphic mythology it is possible to test the optimality of the objective function at the indicated corner points.

Corner Points	Values of the Objective Function [$Z = 60x_1 + 30x_2$]
(0, 0)	Br. 0
(0, 10)	Br. 500
(5, 8)	Br. 700
(9, 4)	Br. 660
(11, 0)	Br. 660

- Thus the optimal maximum values of the given objective function are obtained to be Br. 660; this maximum value is obtained at two different corner points of the feasible solution space, viz. $(11, 0)$ and $(9, 4)$.
- This shows that the given problem has multiple optimal solutions.

2.5. Solution by Simplex Method

- Used when there are more than two decision variables in the problem.
- SIMPLEX METHOD is most powerful method.
- It deals with iterative process, which consists of first designing a
- Basic Feasible Solution or a Programme and proceed towards the OPTIMAL SOLUTION and testing each feasible solution for Optimality to know whether the solution on hand is optimal or not.
- Hence we can say that the Simplex Method depends on two concepts known as Feasibility and optimality.
- Feasibility refers to
- Optimality refers to

- The simplex method is quite simple and mechanical in nature.
- The iterative steps of the simplex method are repeated until a finite optimal solution, if exists, is found.
- If no optimal solution, the method indicates that no finite solution exists.

Comparison Between Graphical And Simplex Methods

1. The graphical method is used when we have two decision variables in the problem. Whereas in Simplex method, the problem may have any number of decision variables.
2. In graphical method, the inequalities are assumed to be equations, so as to enable to draw straight lines. But in Simplex method, the inequalities are converted into equations by:
 - i. Adding a SLACK VARIABLE in maximisation problem and subtracting a SURPLUS VARIABLE in case of minimisation problem.
3. In graphical method, the areas outside the feasible area (area covered by all the lines of constraints in the problem) indicates idle capacity of resource where as in Simplex method, the presence of slack variable indicates the idle capacity of the resources.

Table 2.5.1: General format of a simplex tableau

Basics	C	c_1	c_2	-	c_n	0	0	-	0	Quantity	
		x_1	x_2	-	x_n	s_1	s_2	-	s_m		
s_1	0	a_{11}	a_{12}	-	a_{1n}	1	0	-	0	b_1	
s_2	0	a_{21}	a_{22}	-	a_{2n}	0	1	-	0	b_2	
-	-	-	-	-	-	-	-	-	-		
s_m	0	a_{m1}	a_{m2}	-	a_{mn}	0	0	-	-	b_m	
Z		0	0	-	0	0	0	-	0		
C - Z		c_1	c_2	-	c_n	0	0	-			

BY OSU

C or C_j Row

- Show the coefficients of all variables in the objective function.
- There are two rows needed at the bottom of the tableau in order to complete it.
- The first is a Z row and the second is a C-Z row.
- The entries across the row Z are determined column by column.
- Show the amount of profit/cost the objective function will be reduced by when one unit of the variable, in each column, is brought into the basis.
- For each column, the Z value is obtained by adding the product of each of the numbers in the column and their respective entries in the column C.
- The values in the row C-z are calculated column by column.
- For each column, the value in row Z_j is subtracted from the C_j value in the top row.

Display the net impact on the value of the objective function of bringing one unit of each of the column variable into the basis.

$-c_j$ indicates how much will be gained while z_j shows how much will be simultaneously be lost.

-If the value of $c_j - z_j$ is positive, the value of the objective function can be increased by introducing the variable in that column into the solution.

-In the maximization problems, if one or more of the $c_j - z_j$ values is positive, then the solution is not optimal and it can be improved.

- In the minimization problems, if one or more of the $c_j - z_j$ values is negative, then the solution is not optimal and it can be improved.

Basis Column

- Identify the initial solution variable or basic variables.
- **Unit Profit/Cost Column**
- Display coefficients of basic variables in the objective function.

Quantity Column

- Indicate values of basic variables in the solution.
- For the proposed initial solution, this column displays the right hand side quantities of the constraints.

Columns of Decision, Slack and Surplus Variables

- In the initial tableau, values under each of these columns are coefficients of the corresponding variable in each constraint.
- Each value under these columns shows the rate of utilization of the corresponding resources.
- In other words, it implies how much the value of the relevant basic variable (in the same row) will decrease if the value of the corresponding nonbasic variable increase by 1.

Ratio Column

- This column is used to identify the basic variable that should be removed from the solution (to become a non basic variable) without violating any constraint.
- Each value under this column in each row is the ratio of the Quantity to the coefficient of the variable considered in the same row.

Example~ 1

Maximisation Problem

Example 1. A factory manufactures two products A and B on three machines I, II, and III. Product A requires 10 hours of machine I and 5 hours of machine II and one hour of machine III. The requirement of product B is 6 hours, 10 hours and 2 hours of machine I, II and III respectively. The profit contribution of products A and B are Birr. 23 per unit and Birr. 32 per unit respectively. In the coming planning period the available capacity of machines I, II and III are 2500 hours, 2000 hours and 500 hours respectively.

- Formulate the linear programming problem
- Find the optimal product mix for maximizing the profit using simplex method
- Interpret the result

Solution

The problem can summarised as below table

Machine	Products		Capacity in hours
	A	B	
I	10	6	2500
II	5	10	2000
III	1	2	500
Profit/ Units Birr	23	32	

Step1 Formulate the linear programming problem

Maximize $Z = 23 x_1 + 32 x_2$ S.T. Objective Function

$$10 x_1 + 6 x_2 \leq 2500 \quad \left| \quad \text{Structural Constraints} \right.$$

$$5 x_1 + 10 x_2 \leq 2000$$

$$1 x_1 + 2 x_2 \leq 500$$

And both x_1 and x_2 are ≥ 0 . $\longrightarrow$ Non Negativity

Step 2. Standardize the model. Now the above inequalities are to be converted into equations.

- Introduce the slack variable; slack variable represents the idle capacity represented by S.
- Slack variable for first inequality is S_1 that of second one is S_2 and that of 'n'th inequality is S_n .

Maximize $Z = 23 x_1 + 32 x_2 + 0S_1 + 0S_2 + 0S_3$ S.T.

$$10 x_1 + 6 x_2 + 1S_1 = 2500$$

$$5 x_1 + 10 x_2 + 1S_2 = 2000$$

$$1 x_1 + 2 x_2 + 1S_3 = 500 \text{ and } x_1, x_2, S_1, S_2 \text{ and } S_3 \text{ all } \geq 0.$$

Maximize $Z = 23 x_1 + 32 x_2 + 0S_1 + 0S_2 + 0S_3$ S.T.

$10 x_1 + 6 x_2 + 1S_1 + 0S_2 + 0S_3 = 2500$

$5 x_1 + 6 x_2 + 0S_1 + 1S_2 + 0S_3 = 2000$

$1 x_1 + 2 x_2 + 0S_1 + 0S_2 + 1S_3 = 500$ and x_1, x_2, S_1, S_2 and S_3 all ≥ 0 .

Step 3. Inter the above data in a table known as simplex table (or tableau).

Initial simplex table

Cj 23 32 0 0 0							
Basics	Pro./unit in birr	x ₁	x ₂	s ₁	s ₂	s ₃	Quantity
s ₁	0	10	6	1	0	0	2500/6=416.66
s ₂	0	5	6	0	1	0	2000/6=333.33
s ₃	0	1	2	0	0	1	500/2=250√ (Out going)
Zj		0	0	0	0	0	0

Step 4. Develop the next revised program

For getting the revised program, we have to transfer the rows of table 1 to table 2. To do this the following procedure is used.

Step 1: To Write the incoming variable ' X_2 ' in place of outgoing variable S_2 . Enter the profit of ' X_2 ' in profit column. Do not alter S_1 and S_3 . While doing so DO NOT ALTER THE POSITION OF THE ROWS.

Step 2: DIVIDING THE ELEMENTS OF OLD COLUMN BY KEY COLUMN ELEMENTS obtains capacity column elements.

Step 3: Transfer of key row: DIVIDE ALL ELEMENTS OF KEY ROW BY RESPECTIVE KEY COLUMN NUMBER.

Step 4: Transfer of Non-Key rows: $\text{NEW ROW NUMBER} = (\text{old row number} - \text{corresponding key row number}) \times \text{fixed ratio}$.
 $\text{Fixed ratio} = \text{Key column number of the row} / \text{key number}$.

Step 5: Elements of Net evaluation row are obtained by:

- Objective row element at the top of the row – Σ key column element $\times$ profit column element.
- Step 6: Select the highest positive element in net evaluation row or highest opportunity cost and mark the column by an arrow to indicate key column (incoming variable).
- Step 7: Find the replacement ratios by dividing the capacity column element in the row by key column element of the same row and write the ratios in replacement ratio column. Select the limiting (lowest) ratio and mark with a tick mark to indicate key row (out going variable). The element at the intersection of key column and key row is known as key number.

The element at the intersection of key column and key row is known as key number. Continue these steps until we get:

- (i) For maximization problem all elements of net evaluation row must be either zeros or negative elements.
- (ii) For Minimization problem, the elements of net evaluation row must be either zeros or positive elements.

- X_2 is incoming variable and S_3 is outgoing variable let us develop the next program by undertaking row operation.
- Since 2 is a pivot element divide the key row by 2, to transform the next table.
- $NR_3/2 = 1/2, 1, 0, 0, 1/2 \quad 250$

New row element(NR1) = old row element – corresponding Key row element $\times$ (Key column number/key number).

$$2500 - 500 \times 6/2 = 1000$$

$$10 - 1 \times 6/2 = 7$$

$$6 - 2 \times 6/2 = 0$$

$$1 - 0 \times 6/2 = 1$$

$$0 - 0 \times 6/2 = 0$$

$$0 - 1 \times 6/2 = -3$$

OR, We can compute in the following way

$$NR_1 = -6NR_3 + R_1: -6(1/2) = -3, -6(1) = -6, -6(0) = 0, -6(0) = 0, -6(1/2) = -3, -6(250) = -1500$$

$$-3, -6, 0, 0, -3 / -1500$$

$$+10, 6, 1, 0, 0 / 2500 =$$

$$NR_1 = 7, 0, 1, 0, -3 / 1000$$

**New row element(NR2) = old row element – corresponding
Key row element $\times$ (Key column number/key number).**

$$2000 - 500 \times 6/2 = 500$$

$$5 - 1 \times 6/2 = 2$$

$$6 - 2 \times 6/2 = 0$$

$$0 - 0 \times 6/2 = 0$$

$$1 - 0 \times 6/2 = 1$$

$$0 - 1 \times 6/2 = -3$$

OR It can be computed as follows

$$NR2 = -6NR3 + R2$$

$$\begin{aligned} -6(1/2) &= -3, -6(1) = -6, -6(0) = 0, -6(0) = 0, -6(1/2) = -3, / - \\ 6(250) &= -1500 \end{aligned}$$

$$-3, -6, 0, 0, -3, / -1500$$

$$+5, 6, 0, 1, 0/2000 =$$

$$NR2 = 2, 0, 0, 1, -3 / 500$$

		Cj	23	32	0	0	0		
Basics	Pro./unit in birr	x_1	x_2	s_1	s_2	s_3	Quantity		
s_1	0	7 key	0	1	0	-3	1000/7=142.8√ leaving		
s_2	0	2	0	0	1	-3	500/2=250		
x_2	32	½	1	0	0	½	250/1/2=500		
Zj		16	32	0	0	16	0		
Cj - Zj		7	0	0	0	-16	8000		

S1 is the outgoing variable.

- To transfer key row, divide all the elements of key row by key number (= 7).

$$1000/7 = 142.8, 7/7 = 1, 0/7 = 0, 1/7 = 1/7, 0/7 = 0, 3/7 = \sim 3/7.$$

New row element(NR2) = old row element – corresponding Key row element $\times$ (**Key column number/key number(Fixed ratio)**).

$$500 - 1000 \times 2/7 = 1500/7$$

$$2 - 7 \times 2/7 = 0$$

$$0 - 0 \times 2/7 = 0$$

$$0 - 1 \times 2/7 = \sim 2/7$$

$$1 - 0 \times 2/7 = 1$$

$$\sim 3 - 1 (\sim 3) \times 2/7 = \sim 15/7$$

New row element (NR3) = old row element – corresponding Key row element $\times$ (Key column number/key number).

$$250 - 1000 \times 1/14 = 1250/7$$

$$1/2 - 7 \times 1/14 = 0$$

$$1 - 0 \times 1/14 = 1$$

$$0 - 1 \times 1/14 = -1/14$$

$$0 - 0 \times 1/14 = 0$$

$$1/2 - (-3) \times 1/14 = 5/7$$

Third simplex table

Cj 23 32 0 0 0							
Basics	Pro./unit in birr	x_1	x_2	s_1	s_2	s_3	Quantity
x_1	23	1	0	1/7	0	-3/7	1000/7=142.8
s_2	0	0	0	- 2/7	1	- 15/7	1500/7
x_2	32	0	1	- 1/1 4	0	5/7	1250/7=
Zj		23	32	1	0	13	9000
Cj - Zj		0	0	-1	0	-13	

- Since the net evaluation $C_j - Z_j$ is zero and negative the solution is optimal. Therefore, $X_1=142.85715$ $X_2=178.57143$. The maximum profit is Birr 9000.

Exercise 1. Consider the following LPP and solve using simplex method.

Maximize $Z = 3X_1 + 5X_2 + 4X_3$ s.t.

$$2X_1 + 3X_2 + X_3 \leq 8$$

$$X_1 + 2X_2 + 5X_3 \leq 10$$

$$3X_1 + 2X_2 + 4X_3 \leq 15 \text{ and } X_1, X_2, X_3 \text{ all } \geq 0$$

Minimization Problem

- Above we have discussed how to solve maximization problem and the mechanism or simplex method and interpretation of various elements of rows and columns.
- Now let us see how to solve a minimization problem and see the mechanism of the simplex method in solving and then let us deal with some typical examples so as to make the reader confident to be confident enough to solve problem individually.

S No	Maximization Problem	Minimization Problem
	Similarities	
1.	It has an objective function.	This too has an objective function.
2.	It has structural constraints.	This too has structural constraints.
3.	The relationship between variables and constraints are linear.	Here too the relationship between and variables Constraints are linear.
4.	It has non-negativity constraint	This is too has non-negativity constraint.
5.	The coefficients of variables may be positive or negative or zero.	The coefficient of variables may be positive, Negative or zero.
		BY OSU

	Differences	
1.	The objective function is of maximization type.	The objective function is of minimization type.
2.	The inequalities are of $\leq$ type.	The inequalities are of $\geq$ type.
3.	To convert inequalities into equations, slack variables are added.	To convert inequalities into equations, surplus Variables are subtracted and artificial surplus variables are added.
4.	While selecting incoming variable, highest positive Opportunity cost is selected from net evaluation Row.	While selecting, incoming variable, lowest element in the net evaluation row is selected (highest number with negative sign).
5.	When the elements of net evaluation row are either Negative or zeros, the solution is optimal	When the element of net evaluation row are either Positive or zeros the solution is optimal.

Illustration

The doctor examines the patient and advises him to consume at least 40 units of vitamin A and 50 units of vitamin B daily for a specified time period. He also advises the patient that to get vitamin A and vitamin B he has to drink tonic X and tonic Y that have both vitamin A and vitamin B in a proportion. One unit of tonic X consists 2 units of vitamin A and 3 units of vitamin B and one unit of tonic Y consists of 4 units of vitamin A and 2 units of vitamin B. These tonics are available in medical shops at a cost of Birr 3.00 and Birr 2.50 per unit of X and Y respectively. Now the problem of patient is how much of X and how much of Y is to be purchased from the shop to minimize the total cost and at the same time he can get required amounts of vitamins A and B.

Formulate the linear programming problem

Solve the problem using simplex method

Solution

In this problem, a patient visits the doctor to get treatment for ill health.

Minimize $Z = 3x_1 + 2.5x_2$ S.T

$2x_1 + 4x_2 \geq 40$

$3x_1 + 2x_2 \geq 50$

And both x_1 and x_2 are ≥ 0 .

Standardization

- In the above inequalities, say $2x_1 + 4x_2 \geq 40$, if we give values to x and y such that the sum is greater than or equal to 40,
- For example, $x_1 = 10$ and $x_2 = 10$ then $2x_1 + 4x_2 = 60$ which is > 40 .
- To make it equal to 40 we have to subtract 20, so that $20 + 40 - 20 = 40$.
- When we know the values, we can do this.
- But as we do not know the values of x and y we have to subtract a SURPLUS VARIABLE, generally represented by ' S_1 ', ' S_2 '.... etc.
- If we do this then the inequality $2x_1 + 4x_2 \geq 40$ will be $2x_1 + 4x_2 - 1S_1 = 40$.
- If values of x_1, x_2 are equal to zero, $-1s_1 = 40$, $S_1 = -40$, this violates the principle of non-negativity.
- To overcome this problem we have to introduce artificial surplus A_1 .

- If values of x_1 , x_2 , and S are equal to zero, then $1A_1 = 40$. The artificial surplus variable has the value 40, a positive integer.
- Hence we start our initial programme with the artificial variables, A_1 , A_2 , A_3 etc. and go on replacing them by x_1 , x_2 , x_3 etc. that is decision variables.
- Coming to the **cost coefficients of surplus** and artificial surplus variables, for example, S is **very similar to one unit of S** .
- That is the cost coefficient of ' S ' is zero (it is very much similar to slack variable in maximization problem).
- But the artificial surplus variable has to be purchased by paying a very high price for it.
- The cost coefficient of A_1 is represented by a very high value represented by M (which means one unit of A_1 cost Millions of Birr). As we are introducing CAPITAL ' M ', THIS METHOD IS KNOWN AS BIG ' M ' METHOD.

- Minimize $Z = 3x_1 + 2.5x_2 + 0S_1 + 0S_2 + M A_1 + M A_2$
S.T. Objective Function

$$2x_1 + 4x_2 - 1S_1 + 1A_1 = 40$$

$$3x_1 + 2x_2 - 1S_2 + 1A_2 = 50$$

And $x_1, x_2, S_1, S_2, A_1, A_2$ all ≥ 0 Non negativity Constraint.

Simplex format of the above is:

Minimize $Z = 3x_1 + 2.5x_2 + 0S_1 + 0S_2 + MA_1 + MA_2$ S.T.

$$2x_1 + 4x_2 - 1S_1 + 0S_2 + 1A_1 + 0A_2 = 40$$

$$3x_1 + 2x_2 + 0S_1 - 1S_2 + 0A_1 + 1A_2 = 50$$

And $x_1, x_2, S_1, S_2, A_1, A_2$ all ≥ 0 .

Program Variable	Cost Per Unit	CJ	3 X1	2.5 X2	0 S1	0 S2	M A1	M A2	Cost Requirement
A_1	M		2	4	-1	0	1	0	40
A_2	M		3	2	0	-1	0	1	50
		Zj	5M	6M	-M	-M	M	M	
		Cj-Zj	3-5M	2.5-6M	M	M	0	0	

Note: As the variables A_1 and A_2 are basis variables, their Net evaluation is zero.

Now take 6M and 5M, 6 M is greater and if we subtract 2.5 from that it is negligible. Hence $-6M$ will be the lowest element. The physical interpretation is if patient purchases X_2 now, his cost will be reduced by an amount 6M.

- In other words, if the patient does not purchase the X_2 at this point, his penalty is 6M, i.e., the opportunity cost is 6M.
- As the non-basis variable X_2 has highest opportunity cost (highest element with negative sign), X_2 is the incoming variable.
- Hence, the column under X_2 is key column.
- To find the outgoing variable, divide requirement column element by key column element and find the replacement ratio.
- Select the lowest ratio, i.e., here it is 10, falls in first row, hence
- A_1 is the outgoing variable.

- **To transfer key row, divide all the elements of key row by key number (= 4).**

$$40/4 = 10, 2/4 = 0.5, -1/4 = -0.25, 0/25 = 0, 1/25 = 0.25, 0/4 = 0.$$

New row element = old row element – corresponding Key row element $\times$ (Key column number/key number(Fixed ratio)).

$$50 - 40 \times 2/4 = 30$$

$$3 - 2 \times 0.5 = 2$$

$$2 - 4 \times 0.5 = 0$$

$$0 - (-1) \times 0.5 = 0.5 .$$

$$-1 - 0 \times 0.5 = -1$$

$$0 - 1 \times 0.5 = -0.5$$

$$1 - 0 \times 0.5 = 1$$

Program Variables	Cost Per Unit	CJ	3 X_1	2.5 X_2	0 S_1	0 S_2	M A_1	M A_2	Cost Requirement
X_2	2.5		1/2	1	-1/4	0	1/4	0	10/0.5=20
A_2	M		2	0	1/2	-1	-1/2	1	30= 15
ZJ			1.25+2M	2.5	-0.625 + 0.5M	-M	0.625-0.5M	M	
CJ-ZJ			1.75-2M	0	0.625-0.5M	M	-0.625+1.5M	0	



Changing the key row: $30/2 = 15$, $2/2 = 1$, $0/2 = 0$, $0.5/2 = 0.25$, $-1/2 = -0.5$, $-0.5/2 = -0.25$, $1/2 = 0.5$.

Changing the non key row:

$$10 - 30 \times 0.5/2 = 2.5$$

$$0.5 - 2 \times 0.25 = 0$$

$$1 - 0 \times 0.25 = 1$$

$$-0.25 - 0.5 \times 0.25 = -0.375$$

Program Variable	Cost Per Unit	CJ	3 X_1	2.5 X_2	0 S_1	0 S_2	M A_1	M A_2	Cost Requirement
X_2	2.5		0	1	$-3/8$	$1/4$	$3/8$	$-1/4$	2.5
X_1	3		1	0	$1/4$	$-1/2$	$-1/4$	$1/2$	15
Z_J			3	2.5	$-3/16$	$-7/8$	$3/16$	$7/8$	
$C_J - Z_J$			0	0	$3/16$	$7/8$	$M - 3/16$	$M - 7/8$	

- **As all the elements of net evaluation row are either zeros or positive elements the solution is optimal.**
- **$X_1 = 15, X_2 = 2.5$**
- **Optimal Cost = $Z^* = 3 \times 15 + 2.5 \times 2.5 = 45 + 6.25 = \text{Birr. } 51.25$**
- **Imputed value = $0.1875 \times 40 + 0.875 \times 50 = 7.5 + 43.75 = \text{Birr. } 51.25$.**
- **The patient has to purchase 15 units of X_1 and 2.5 units of X_2 to meet the requirement and the cost is Birr 51.25**

Minimize $Z = 5x_1 + 4x_2 + 3x_3$

Subject to the constraints

$$x_1 + x_2 + x_3 \geq 100$$

$$2x_1 + x_2 \geq 50$$

All variables ≥ 0

- a. Solve the problem using Big Method**
- b. Interpret the results**

2.6. Special Cases

1. Unbounded Solutions

- A solution is unbounded if:
 - ✓ The objective function can be improved without limit.
 - ✓ Arises when there are no positive ratios(Replacement ratio).
 - ✓ A zero ratio means that increasing the variable which is attached with that negative ratio would not use any resource.
 - ✓ It occurs when objective function can be improved without limit.
 - ✓ Occurs when there are no positive ratios in process of selecting leaving basis variable based no replacement ratio.

Example 2.5.1:

Maximize

- Subject to the constraints

$$x_1 - x_2 \leq 10$$

- $2x_1 - x_2 \leq 40$
All variables ≥ 0

2. Degeneracy

- Degeneracy occurred When subsequent iteration shows limitation of existing rule to proceed further.
- It might occur right in the initial simplex tableau
- It happens when the number of constraints is less than the number of decision variables in the objective function.
- The problem can be overcome by trial and error method.

Example

$$Z = 2x_1 + 3x_2 + 10x_3$$

$$x_1 + 2x_3 \leq 0$$

$$x_2 + x_3 \leq 3$$

$$x_1, x_2 \geq 0$$

3. Multiple Optimal Solutions

- When all the non basic variables have zero values across the C~Z row of the final simplex table, it can be considered as a signal for the existence of a multiple optimal solution.

Example

$$\begin{array}{ll}\text{Max} & Z = 6x_1 + 4x_2 \\ & -2x_1 + x_2 \leq 2 \\ & 3x_1 + 2x_2 \leq 9 \\ & \text{All variables} \geq 0\end{array}$$

4. Infeasibility

- This is the case in which we get minimum or maximum value of the objective function even before all the decision variables are taking a placement in the basic variables column Basics.
- The optimal solutions obtained at such a situation are called **Optimum Basic Infeasible Solutions**.
- Note that in a big m method infeasibility is said to exist when artificial variables presented in the final simplex tables across the basic variable column.

Chapter Three

Post Optimality Analysis and Duality in LPP

3.1.Sensitivity Analysis

Introduction

- While solving a linear programming problem for optimal solution, we assume that: (a). Technology is fixed, (b). Fixed prices, (c). Fixed levels of resources or requirements, (d). The coefficients of variables in structural constraints (i.e. time required by a product on a particular resource) are fixed, and (e) profit contribution of the product will not vary during the planning period.
- These assumptions, implying certainty, complete knowledge, and static conditions, permit us to design an optimal programme.
- The condition in the real world however, might be different from those that are assumed by the model.
- It is, therefore, desirable to determine how sensitive the optimal solution is to different types of changes in the problem data and parameters.

1.1.A change in the Right Hand Side [RHS] Values of a Constraint

- Determining how a change in the RHS values of a constraint that would influence the optimal solution

Step 1. Examine the shadow prices in the final simplex tableau of the optimality analysis.

- What is shadow prices?
- Shadow prices are the coefficients of the slack variable in the Z row of the final simplex tableau.
- Shadow prices is a marginal value; it indicates the impact that a one unit change in the RHS value of a problem constraint would have on the value of the objective function.
- Shadow price reveals the amount by which the value of the objective function would increase if the level of the RHS value of a given problem constraint was increased by one unit.

- shadow price do not tell us the extents to which the level of a scarce resource can be changed and still have the same impact per unit.
- We need to determine the range over which we can change the RHS quantity of a problem constraint given that there is the same shadow price.
- This range is called Range of Feasibility or the Right Hand Side Range.

Step 2 Construct a range of feasibility for each and every RHS values of the existing constraints of the LP model.

- The key to computing the range of feasibility for the constraints lies in the final simplex table of the optimal analysis.
- Purposely, the entries in the body of the final simplex tableau in each slack column must be used to compute the range of feasibility for the corresponding constraint.

In forming the range we need to consider the following.

- The entries in the associated slack column should be identified.
- Ratios of current values to corresponding entries of the slack column should be determined. This would be carried out when the entries in the associated slack column have been divided into the corresponding values in the current value column.

The lower and upper limits of the range should be determined using.

For a maximization problem

- Lower limit = current level of the constraint minus the positive ratio nearer to zero.
- Upper limit = current level of the constraint minus the negative ratio nearer to zero.

For a minimization problem

- Lower limit = current level of the constraint minus the negative ratio nearer to zero.
- Upper Limit = current level of the constraint minus the positive ratio nearer to zero.

Example 1.1.1:

Given below is the LP model and the associated final simplex tableau of the optimality analysis

Table 1.1.1. Identify the shadow prices and determine the range of feasibility.

Maximize $Z = 3x_1 + 4x_2$

1. $3x_1 + 5x_2 \leq 15$

2. $2x_1 + x_2 \leq 8$

3. $x_2 \leq 2$

All variables ≥ 0

Example

Table 1.1.1: Final simplex tableau

Basics	C	3	4	0	0	0	Quantity
		x_1	x_2	s_1	s_2	s_3	
x_1	3	1	0	-0.143	0.714	0	3.57
s_3	0	0	0	-0.286	0.428	1	1.143
x_2	4	0	1	0.286	-0.428	0	0.857
Z		3	4	0.714	0.428	0	14.138
C - Z		0	0	-0.714	-0.428	0	

Solution:

- The shadow prices are entries along row Z which fall under the three slack variables , and .
- These values are 0.714, 0.428, and 0, respectively.
- 0.714 is marginal value; it indicates the impact that a one unit change in the RHS value [15] of the first problem constraint would have on the value of the objective function.
- That is, the value of the objective function [14.138] would be increased into $14.138 + 0.714 = 14.852$ if the level of RHS value of the first problem constraint is increased by one unit.

Exercise

Analysis on RHS of the second and third constraints.

1.2. A Change in an Objective Function Coefficient.

- It is useful to know how much the contribution of a given decision variable to the objective function can change without changing the optimal solution.
- There are two cases to consider:
 1. Changes for the coefficient of a decision variable that is not currently in the solution mix [Non Basic Decision Variable],
 2. Changes for the coefficient of decision variable that is currently in the solution mix [Basic Decision Variable].

1. Changes for the coefficient of a decision variable that is not currently in the solution mix [Non Basic Decision Variable]

Range of Insignificance

- If a decision variable is not currently in solution mix in maximization problem, its coefficient in the objective function would have to increase by an amount that exceeds its corresponding value along the Z row in the final simplex tableau in order for it to end up as a basic variable in the optimal solution mix.

- That is, in order for the variable to be included in the optimal solution, its coefficient in the objective function will have to change from the existing coefficient to a new coefficient level.
- The new coefficient level should always exceed the corresponding value of the non basic decision variable along the row Z.

- Hence, the range over which the coefficient of a non basic decision variable in the objective function can change without causing the optimal solutions of the basic decision variables is said to be Range of Insignificance.
- The upper limit of such arrange is given in the Z~row under the non basic decision variable column, which is $0 \leq C_j[\text{new}] \leq Z_j$
- where
- $C_j [\text{new}]$ is the new coefficient of the jth non basic decision variable, and
- Z_j is the corresponding value of the jth non basic decision variable column along the Z row.

Example

Given the final simplex tableau, determine the range over which the coefficient of the non basic decision variable in the objective function could change without changing the optimal solution mix.

Basics	C	5	4.5	1	0	0	0	Quantity
		x_1	x_2	x_3	s_1	s_2	s_3	
x_1	5	1	1.053	0	0.067	0	0	10
x_3	1	0	0.67	1	-0.022	0.067	0	1.8
s_3	0	0	1.924	0	0.258	-0.773	1	15.12
Z		5	5.342	1	0.311	0.067	0	51.80
C – Z		0	-0.842	0	-0.311	-0.067	0	

Solution

- Because X_2 is not in the solution mix, its coefficient in the objective function would need to increase in order for it to come into solution.
- The amount of increase must be greater than its corresponding value along the row C~Z, which - 0.842.
- Therefore, its existing coefficient in the objective function must be increase by more than Br.0.842.
- Because its current value is Br. 4.5, the current solution mix will remain optimal as long as its coefficient in the objective function does not exceed $\text{Br. } 4.5 + \text{Br. } 0.842$, or Br. 5.342.

- For the reason that an increase would be needed to bring it into the solution mix, any lesser value than Br. 5.342 of its coefficient in the objective function would keep it out of the solution mix.
- Hence, the range of insignificance for X_2 is Br. 5.342 or less.
- That is $\sim <C_2[\text{new}] < 5.342$.
- Note, however, that, in order for the variable to be included in the optimal solution, its coefficient in the objective function will have to change from the existing coefficient Br.4.5 into the new level, which should exceed the corresponding value of this variable along the row ZBr.5.342.

2. Changes for the coefficient of decision variable that is currently in the solution mix [Basic Decision Variable].

Range of Optimality

- The range over which the coefficient of a basic decision variable can change without shifting the optimal solutions of the basic decision variables is called its Range of Optimality.

- For variables which are in the solution mix:
 - the determination of the range of optimality requires the entries along the row $C - Z$ and the corresponding row values of the basic decision variable in question in the body of the final simplex tableau.
 - In order to form the range of optimality one has to carefully consider the following events:
 - The entries along the row $C - Z$ should be identified;
 - The values in row $C - Z$ must be divided by the corresponding row values of the basic decision variable in question.
 - They will result in a series of ratios.

- The positive ratio closest to zero will indicate the amount of allowable increase to the current value and the negative ratio closest to zero will indicate the amount of allowable decrease from the current value and
- Finally, the lower and upper limits of the range should be determined accordingly this rule applies for both Maximization and Minimization problem.
- Lower Limit = Existing coefficient of the basic decision variable in question plus the negative ratio nearer to zero.
- Upper Limit = Existing coefficient of the basic decision variable in question plus the positive ratio nearer to zero.

Example

Determine the range of optimality for the decision variables in the solution mix.

$$\text{Maximize } Z = 12x_1 + 10x_2 + 15x_3 + 11x_4$$

Subject to:

$$\text{A: } 5x_1 + 3x_2 + 4x_3 + 2x_4 \leq 240 \quad \text{pounds [Material]}$$

$$\text{B: } 6x_1 + 8x_2 + 2x_3 + 3x_4 \leq 240 \quad \text{hours [Machine]}$$

$$\text{C: } 2x_1 + 3x_2 + 3x_3 + 2x_4 \leq 180 \quad \text{hours [Labor]}$$

All variables ≥ 0

Optimal simplex tableau of the firm

Basics	C	12	10	15	11	0	0	0	Quantity
		x_1	x_2	x_3	x_4	s_1	s_2	s_3	
x_3	15	-1.2	-1.4	1	0	0	-0.4	0.6	12
s_1	0	4.2	1.4	0	0	1	0.4	-1.6	48
x_4	11	2.86	3.6	0	1	0	0.6	-0.4	72
Z		12.8	18.6	15	11	0	0.6	4.6	972
C - Z		-0.8	-8.6	0	0	0	-0.6	-4.6	

Solution:

- For variables which are in the solution mix, the determination of the range of optimality requires the entries along the row $C - Z$ and the corresponding row values of the basic decision variable in question in the body of the final simplex tableau.

Analysis on the coefficient of x_3 in the objective function:

	x_1	x_2	x_3	x_4	s_1	s_2	s_3
C - Z	-0.8	-8.6	0	0	0	-0.6	-4.6
x_3	-1.2	-1.4	1	0	0	-0.4	0.6
$C - Z/x_3$	0.67	6.14	0	Undefined	Undefined	1.5	-7.67

- The lower and upper limit of the range of optimality over which coefficient of the basic decision variable x_3 can change is obtained to be:
- Lower Limit = $15 + [-7.67] = 7.33$. This indicates the coefficient of can be decreased by Br. 7.67 before it hits the lower limit of the range of optimality.
- Upper Limit = $15 + [0.67] = 15.67$. This indicates the coefficient of can be increased by Br. 0.67 before it reaches the upper limit of the range of optimality.
- Therefore, $7.33 \leq \leq 15.67$ is said to be the range of optimality over which the profit per unit of product three is valid.

Analysis on the coefficient of x_4 in the objective function:

	x_1	x_2	x_3	x_4	s_1	s_2	s_3
C - Z	-0.8	-8.6	0	0	0	-0.6	-4.6
x_4	2.8	3.6	0	1	0	-0.6	-0.4
C - Z/x_4	-0.29	-2.39	Undefined	0	Undefined	-1	11.5

- The lower and upper limit of the range of optimality for which the coefficient of the basic decision variable in an objective function can change is found to be:
- Lower Limit = $11 + [-0.29] = 10.71$. This indicates the coefficient of can be decreased by Br. 0.29 before it hits the lower limit of the range of optimality.
- Upper Limit = $11 + [11.5] = 22.5$. This indicates the coefficient of can be increased by Br. 11.5 before it reaches the upper limit of the range of optimality.

Exercise

Given this LP model and its final simplex tableau

Maximize $10 x_1 + 6 x_2 + 5 x_3$

Subject to

$$A: 2 x_1 + 3 x_2 + 4 x_3 \leq 25$$

$$B: x_1 + 3x_2 + 2 x_3 \leq 22$$

$$C: 6 x_1 + 3x_2 + 4x_3 \leq 32$$

All the decision variables are non zero

Optimal simplex tableau of the firm

Basics	C	10	6	5	0	0	0	Quantity
		x_1	x_2	x_3	s_1	s_2	s_3	
s_1	0	0	0	$8/5$	1	$-4/5$	$-1/5$	1
x_2	6	0	1	$8/15$	0	$2/5$	$-1/15$	$20/3$
x_1	10	1	0	$2/5$	0	$-1/5$	$1/5$	2
Z		10	6	$36/5$	0	$2/5$	$8/5$	60
C - Z		0	0	$-11/5$	0	$-2/5$	$-8/5$	

- Identify the shadow prices and indicate the constraints with which each shadow price is associated and interpret any one of the shadow prices.
- Determine the range of feasibility for each b_i [That is, original right hand side value].
- Determine range of optimality for the decision variables that are in solution.
- By how much would the coefficient of x_3 in the objective function have to increase before it would come into solution? What is the range of insignificance for x_3 ?
- If one additional unit of the third resource [constraint] could be obtained at a premium of Br. 0.50, what impact would this have on profit?

Duality in LPP

- Linear programming problems exist in pairs.
- That is in linear programming problem, every maximization problem is associated with a minimization problem.
- Conversely, associated with every minimization problem is a maximization problem.
- Once we have a problem with its objective function as maximization, we can write by using duality relationship of linear programming problems, its minimization version.
- The original linear programming problem is known as primal problem, and the derived problem is known as dual problem.

Primal problem:

Minimize $Z = 3x_1 + 2.5x_2$. s.t.

$$2x_1 + 4x_2 \geq 40$$

$$3x_1 + 2x_2 \geq 50$$

$$x_1, x_2 \geq 0$$

Dual Problem

Maximize $Z = 40y_1 + 50y_2$. S.t

$$2y_1 + 3y_2 \leq 3$$

$$4y_1 + 2y_2 \leq 2.5$$

$$y_1, y_2 \geq 0$$

Chapter Four

Transportation and Assignment Problems

Transportation Problems

- Transportation problem is special case of a Linear Programming Problem.
- The objective is to transport a homogeneous commodity from various origins or factories to different destinations or markets at a total minimum cost.

The properties of linear programming.

1. It has an objective function.
2. It has structural constraints.
3. It has a non-negativity constraint.
4. The relationship between the variables and the constraints are linear.

Formulation of the Transportation Models

- In order to design a transportation technique one needs to have the following fundamental information, namely:
- Maximum availability of quantity [capacity] at various origins;
- Maximum requirements of quantity [order] at different destination; and
- Cost or profit of transportation of a unit item from each origin to every destination.

Approach To Solution To A Transportation Problem By Using Transportation Algorithm

Methods for Finding Initial Solution

A .Basic feasible solution can be obtained by Following Methods:

- (a) North ~ west corner method.
- (b) Row minimum
- (c) Column minimum method
- (d) Least ~ cost cell method or Inspection method Or Matrix minimum
- (e) Vogel's Approximation Method, generally known as VAM.

Example

HHH Company has 3 factories A,B and C which supply items to retailers X, Y and Z. Monthly factory capacities are 50, 15 and 55 at Factory A, Factory B, and Factory C, respectively. Monthly retailers requirements at X, Y and Z are 25, 50 and 45, respectively. Unit item transportation costs in Birr are given below table.

To From	Retailer X	Retailer Y	Retailer Z	Supply Quantity
Factory A	Br. 12	Br. 20	Br. 15	50
Factory B	Br. 9	Br. 11	Br. 4	15
Factory C	Br. 20	Br. 14	Br. 8	55
Demand Quantity	25	50	45	120

a. Find an initial feasible solution to the transportation problem given above in the HHH Company using North - west corner method, using North - west corner method.

Solution

It is very simple and easy to understand.

Steps

1. Make the first assignment to the upper left-hand cell which is called North-West corner of the transportation table.
2. The quantity allocated should be the lesser of the row supply and the column demand.
3. Amend the row and column totals accordingly.
2. Remain in a row or column until its supply or demand is completely allocated.
3. Assign the maximum number of units to every cell in turn until all supply is allocated and all demand has been satisfied.

Initial feasible solution Using North west Method

From \ To	Retailer X	Retailer Y	Retailer Z	Supply Quantity
Factory A	Br. 12 25 [1 st]	Br. 20 25 [2 nd]	Br. 15	50
Factory B	Br. 9	Br. 11 15 [3 rd]	Br. 4	15
Factory C	Br. 20	Br. 14 10 [4 th]	Br. 8 45 [5 th]	55
Demand Quantity	25	50	45	120

- Multiplying the quantities in the completed cells by the corresponding cell's unit transportation costs and then adding them will yield the total transportation cost of the problem. That is, Total Transportation Cost, T is given by:

$$T = 12(25) + 20(25) + 11(15) + 14(10) + 8(45) = \text{Br. } 1465$$

b) Find an initial feasible solution to the transportation problem given above in the HHH Company using **~Row – Minimum Rule Method**.

Solution

The following set of principles guides the allocations:

1. Locate a cell with the smallest unit transportation cost in the first row and allocate a quantity that is equal to the lower of the available row supply and column demand;
2. Delete or cross out the cells in the row or column that has been exhausted [or both, if both have been exhausted], and adjust the remaining row or column total accordingly; and
3. Repeat 1 and 2 till all allocations are made.

To From	Retailer X	Retailer Y	Retailer Z	Supply Quantity
Factory A	Br. 12 25 [1 st]	Br. 20	Br. 15 25 [2 nd]	50
Factory B	Br. 9	Br. 11	Br. 4 15 [3 rd]	15
Factory C	Br. 20	Br. 14 50 [5 th]	Br. 8 5 [4 th]	55
Demand Quantity	25	50	45	120

Thus, the Transportation Cost, T is given by:

$$\begin{aligned}
 T &= 12(25) + 15(25) + 4(15) + 14(50) + 8(5) = \\
 &= \text{Br. } 1475
 \end{aligned}$$

c) Find an initial feasible solution to the transportation problem given above in the HHH Company using ~Column – Minimum Rule Method.

Solution

Steps

The following set of principles guides the allocations:

1. Locate a cell with the smallest unit transportation cost in the first column and allocate a quantity that is equal to the lower of the available row supply and column demand;
2. Delete or cross out the cells in the row or column that has been exhausted [or bot, if both have been exhausted], and adjust the remaining row or column total accordingly; and
3. Repeat 1 and 2 still all allocations are made.

To From	Retailer X	Retailer Y	Retailer Z	Supply Quantity
Factory A	Br. 12 25 [1 st]	Br. 20	Br. 15 25 [5 th]	50
Factory B	Br. 9	Br. 11 15 [2 nd]	Br. 4	15
Factory C	Br. 20	Br. 14 35 [3 rd]	Br. 8 20 [4 th]	55
Demand Quantity	25	50	45	120

Thus, the Transportation Cost, T is given by:

$$T = 12(10) + 15(9) + 14(50) + 40(15) + 5(8)$$

$$= \text{Br.}$$

d. Table Minimum – Rule (Method) [Intuitive Method]

- This method also known as minimum cost method. The method uses the cell with the lowest none zero unit cost in the entire table as a basis for allocating quantities.

The following set of principles guide the allocations:

1. Locate a cell with the smallest unit transportation cost in the table and assign a quantity that is equal to the lower of the available row supply and column demand;
2. Delete or cross out the cells in the row or column that has been exhausted [or both, if both have been exhausted], and adjust the remaining row or column total accordingly;
3. Repeat 1 and 2 till all allocations are made.

Solution

To From	Retailer X	Retailer Y	Retailer Z	Supply Quantity
Factory A	Br. 12 25 [3 rd]	Br. 20 25 [5 th]	Br. 15	50
Factory B	Br. 9	Br. 11	Br. 4 15 [1 st]	15
Factory C	Br. 20	Br. 14 25 [4 th]	Br. 8 30 [2 nd]	55
Demand Quantity	25	50	45	120

Thus, the total Transportation Cost, T is given by:

$$\begin{aligned}
 T &= 12(25) + 20(25) + 4(15) + 14(25) + 8(30) \\
 &= \text{Br. } 1450
 \end{aligned}$$

5. Solution by Vogel's Approximation Method: (Opportunity cost method) : Also known as penalty Rule method

- In this method, we use concept of **opportunity cost**.
Opportunity cost is the penalty for not taking correct decision.

Steps

- i. Compute the row and column opportunity costs
 - *To find the row opportunity cost in the given matrix deduct the smallest element in the row from the next highest element.*
 - *To calculate the column opportunity cost, deduct smallest element in the column from the next highest element.*
- *Write row opportunity costs of each row just by the side of availability constraint and similarly write the column opportunity cost of each column just below the requirement constraints.*
- *These are known as penalty column and penalty row.*

- (ii) Write row opportunity costs and column opportunity costs as described above.
- (iii) Identify the highest opportunity cost among all the opportunity costs and write a tick ($\checkmark$) mark at that element.
- (iv) If there are two or more of the opportunity costs which of same magnitude, then select any one of them, to break the tie.

e. Find an initial feasible solution to the transportation problem given above in the HHH Company using VAM.

From \ To	Retailer X	Retailer Y	Retailer Z	Supply Quantity
Factory A	Br. 12	Br. 20	Br. 15	50
Factory B	Br. 9	Br. 11	Br. 4	15
Factory C	Br. 20	Br. 14	Br. 8	55
Demand Quantity	25	50	45	120

	X	Y	Z	SS
A	12 25	20 25	15	50
B	9	11 15	4	15
C	20	14 10	8 45	55
DD	25	50	45	120

- Total cost by VAM

$$T = 25 \times 12 + 25 \times 20 + 15 \times 11 + 10 \times 14 + 45 \times 8 = \text{Birr } 1465$$

Optimality Test: (Approach to Optimal Solution)

- Once, we get the basic feasible solution for a transportation problem, the next duty is to test whether the solution got is an optimal one or not? This can be done by two methods.

(a) By Stepping Stone Method, and

(b) By Modified Distribution Method, or MODI method.

BN. An optimality test can however be performed on the feasible solution in which number of allocation is $m+n-1$.

(a) Stepping stone method of optimality test

- To give an optimality test to the solution obtained, we have to find the opportunity cost of empty cells.
- As the transportation problem involves decision making under certainty, we know that an optimal solution must not incur any positive opportunity cost.
- Thus, we have to determine whether any positive opportunity cost is associated with a given programme, i.e., for empty cells.
- **Once the opportunity cost of all empty cells are negative, the solution is said to be optimal.**
- In case any one cell has got positive opportunity cost, then the solution is to be modified. The Stepping stone method is used for finding the opportunity costs of empty cells. Every empty cell is to be evaluated for its opportunity cost. To do this the methodology is:

1. Put a small '+' mark in the empty cell.
2. Starting from that cell draw a loop moving horizontally and vertically from loaded cell to loaded cell.
 - Remember, there should not be any diagonal movement. We have to take turn only at loaded cells and move to vertically downward or upward or horizontally to reach another loaded cell.
 - In between, if we have a loaded cell, where we cannot take a turn, ignore that and proceed to next loaded cell in that row or column.
3. After completing the loop, mark minus (−) and plus (+) signs alternatively.
4. Identify the lowest load in the cells marked with negative sign.

5. This number is to be added to the cells where plus sign is marked and subtract from the load of the cell where negative sign is marked.
6. Do not alter the loaded cells, which are not in the loop.
7. The process of adding and subtracting at each turn or corner is necessary to see that rim requirements are satisfied.
8. Construct a table of empty cells and work out the cost change for a shift of load from loaded cell to loaded cell.

9. If the cost change is positive, it means that if we include the evaluated cell in the programme, the cost will increase. If the cost change is negative, the total cost will decrease, by including the evaluated cell in the programme.

10. The negative of cost change is the opportunity cost. Hence, in the optimal solution of transportation problem empty cells should not have positive opportunity cost.

11. Once all the empty cells have negative opportunity cost, the solution is said to be optimal.

Example

Test the for optimality of Initial Feasible Solution for HHH Co. obtained using the North – West Corner method using stepping stone method.

To From	Retailer X	Retailer Y	Retailer Z	Supply Quantity
Factory A	Br. 12 25 [1 st]	Br. 20 25 [2 nd]	Br. 15	50
Factory B	Br. 9	Br. 11 15 [3 rd]	Br. 4	15
Factory C	Br. 20	Br. 14 10 [4 th]	Br. 8 45 [5 th]	55
Demand Quantity	25	50	45	120

MODI (Modified Distribution method)

- In MODI (Modified Distribution method), we can get the opportunity costs of empty cells without writing the loop. After getting the opportunity cost of all the cells, we have to select the cell with highest positive opportunity cost for including it in the modified solution.

Steps in MODI method:

1. Select row element (u_i) and Column element (v_j) for each row and column, such that $u_i + v_j =$ the actual cost of loaded cell.
2. In MODI method we can evaluate empty cells simultaneously and get the opportunity cost of the cell by using the formula $(u_i + v_j) - C_{ij}$, where C_{ij} is the actual cost of the cell.

2. In resource allocation problem (maximization or minimization method), we have seen that once any variable becomes basis variable, *i.e.*, the variable enters the programme; its opportunity cost or net evaluation will be zero.

Here, in transportation problem also, once any cell is loaded, its opportunity cost will be zero. Now the opportunity cost is given by $(u_i + v_j) - C_{ij}$, which is, equals to zero for a loaded cell.

i.e. $(u_i + v_j) - C_{ij} = 0$ which means, $(u_i + v_j) = C_{ij}$. Here $(u_i + v_j)$ is known as **implied cost of the cell**.

For any loaded cell the implied cost is equals to actual cost of the cell as its opportunity cost is zero.

For any empty cell, (implied cost – actual cost) will give opportunity cost.

3. How to select u_i and v_j ? The answer is:

- (a) Write arbitrarily any one of them against a row or against a column. The written u_i or v_j may be any whole number *i.e* u_i or v_j may be $\leq$ or $\geq$ to zero.

By using the formula $(u_i + v_j) = C_{ij}$ for a loaded cell, we can write the other row or column element.

For example, if the actual cost of the cell $C_{ij} = 5$ and arbitrarily we have selected $u_i = 0$, then v_j is given by $u_i + v_j = 0 + v_j = 5$. Hence $v_j = -5$.

Like this, we can go from loaded cell to loaded cell and complete entering of all u_i s and v_j s.

- (b) Once we get all u_i s and v_j s, we can evaluate empty cells by using the formula $(u_i + v_j) - \text{Actual cost of the cell} = \text{opportunity cost of the cell}$, and write the opportunity cost of each empty cell at left hand bottom corner.

- **Once the opportunity costs of all empty cells are negative, the solution is said to be optimal.** In case any cell is having the positive opportunity cost, the programme is to be modified.
- Remember the formula that **IMPLIED COST OF A CELL** = $u_i + v_j$ Opportunity cost of loaded cell is zero *i.e* $(u_i + v_j) = \text{Actual cost of the cell}$.
- Opportunity cost of an empty cell = implied cost – actual cost of the cell = $(u_i + v_j) - C_{ij}$

Example test for optimality using MODI Method.

To From	Retailer X	Retailer Y	Retailer Z	Supply Quantity
Factory A	Br. 12 25 [1 st]	Br. 20 25 [2 nd]	Br. 15	50
Factory B	Br. 9	Br. 11 15 [3 rd]	Br. 4	15
Factory C	Br. 20	Br. 14 10 [4 th]	Br. 8 45 [5 th]	55
Demand Quantity	25	50	45	120

Assignment Problems

- The assignment problem is a special type of linear programming problem where assignees are being assigned to perform tasks.
- Assigning people to jobs is a common application of the assignment problem.
- However, the assignees need not be people.
- They also could be machines, or vehicles, or plants, or even time slots to be assigned tasks.
- The first example below involves operators being assigned to bulldozers, so the tasks in this case simply involve operating a machine.

- To fit the definition of an assignment problem, these kinds of applications need to be formulated in a way that satisfies the following assumptions.
 1. The number of assignees and the number of tasks are the same. (This number is denoted by n .)
 2. Each assignee is to be assigned to exactly *one* task.
 3. Each task is to be performed by exactly *one* assignee.
 4. There is a cost c_{ij} associated with assignee i ($i = 1, 2, \dots, n$) performing task j ($j = 1, 2, \dots, n$).
 5. The objective is to determine how all n assignments should be made to minimize the total cost.

Basic Assumption in Assignment Problems

- Conditions for a problem to be considered as an assignment problem:
- Number of items in one group is equal to number of items in the other group in the one for one matching;
- Each item of the first group is expected to be assigned to one and only one item of the second group;
- Each item [party] of the first group is independently capable enough to be matched to any one of the elements in the other group;
- Assignment criteria should be clearly stated in such a way that they fit with the objective function of the problem; and
- The cost or profit at each and every possible assignment should be clearly specified.

Approach to Solution

Hungarian Method / Flood's technique / Assignment algorithm: (opportunity cost method)

Steps

Step 1. Deduct the smallest element in each row from the other elements of the row.

- The matrix thus got is known as Row opportunity cost matrix (ROCM).
- The logic here is if we assign the job to any machine having higher cost or time, then we have to bear the penalty.
- If we subtract smallest element in the row or from all other element of the row, there will be at least one cell having zero, i.e zero opportunity cost or zero penalty.
- Hence that cell is more competent one for assignment.

Step 2. Deduct the smallest element in each column from other elements of the column.

- The matrix thus got is known as **Column opportunity cost matrix (COCM)**.
- Here also by creating a zero by subtracting smallest element from all other elements we can see the penalty that one has to bear.
- Zero opportunity cell is more competent for assignment.

Step 3. Add COCM and ROCM to get the **Total opportunity cost matrix (TOCM)**.

Step 4. (modified): Total opportunity cost matrix can be got by simplify doing row operation on

- Column opportunity matrix or column operation on row opportunity cost matrix. This method is simple one and saves time. (Doing row operation on column opportunity matrix means:
- Deduct the smallest element in the row from all other elements in the row in column opportunity matrix and vice versa).
- The property of total opportunity cost matrix is that it will have at least one zero in every row and column. All the cells, which have zero as the opportunity cost, are eligible for assignment.

Step 5. Once we get the total opportunity cost matrix, cover all the zeros by **MINIMUM NUMBER OF HORIZONTAL AND VERTICAL LINES**. (First cover row or column, which is having maximum number of zeros and then next row or column having next highest number of zeros and so on until all zeros are covered.

- Remember, only horizontal and vertical lines are to be drawn.

Step 6. If the lines thus drawn are equal to the number of rows or columns (because of square matrix), we can make assignment. If lines drawn are not equal to the number of rows or columns go to step 7.

Step 7. To make assignment: Search for a single zero either row wise or column wise. If you start row wise, proceed row by row in search of single zero. Once you find a single zero; assign that cell by enclosing the element of the cell by a **square**. Once all the rows are over, then start column wise and once you find single zero assign that cell and enclose the element of the one cell in a square. Once the assignment is made, then all the zeros in the row and column corresponding to the assigned cell should be cancelled. Continue this procedure until all assignments are made. Some times we may not find single zero and find more than one zero in a row or column. It indicates, that the problem has an alternate solution. We can write alternate solutions. (The situation is known as a **TIE** in assignment problem).

- **Step 8.** If the lines drawn are less than the number of rows or columns, then we cannot make assignment. Hence the following procedure is to be followed:
 - The cells covered by the lines are known as **Covered cells**. The cells, which are not covered by lines, are known as **uncovered cells**. The cells at the intersection of horizontal line and vertical lines are known as **Crossed cells**.
- (a) Identify the smallest element in the uncovered cells.
- (i) Subtract this element from the elements of all other uncovered cells.
- (ii) Add this element to the elements of the crossed cells.
- (iii) Do not alter the elements of covered cells.

(b) Once again cover all the zeros by **minimum number of horizontal and vertical lines**.

(c) Once the lines drawn are equal to the number of rows or columns, assignment can be made as said in step (6).

(d) If the lines are not equal to number of rows or columns, repeat the steps 7 (a) and 7 (b) until we get the number of horizontal and vertical lines drawn are equal to the number of rows or columns and make allocations as explained in step (6).

Example

- A manager may be faced with the task of assigning four operators to four bulldozers, one operator for one bulldozer. A manager has prepared a table that shows the operation cost of each four bulldozers by each of the four operators as shown below. According to the following Bulldozer 1 will cost Br. 70 if it is operated by Operator A, and so on.
- The manager has stated that his objective is to develop a set of operator assignment that will minimize the total cost of getting all bulldozers operated. It is further required that the bulldozers be operated simultaneously.

	Bulldozer			
	1	2	3	4
Operator A	Br. 70	Br. 80	Br. 75	Br. 64
Operator B	Br. 55	Br. 52	Br. 58	Br. 54
Operator C	Br. 58	Br. 56	Br. 64	Br. 68
Operator D	Br. 62	Br. 60	Br. 67	Br. 70

Chapter Four

Decision Theory

WHAT IS A DECISION?

- A **decision** is the conclusion of a process designed to weigh the relative utilities or merits of a set of available alternatives so that the most preferred course of action can be selected for implementation.
- Decision-making involves all that is necessary to identify the most preferred choice to satisfy the desired goal or objective.
- Hence decision-making process must involve a set of goals or objectives, a system of priorities, methods of enumerating the alternative courses of feasible and viable courses and a system of identifying the most favourable alternative.

- One must remember that the decisions are sequential in nature.
- It means to say that once we select an alternative, immediately another question arises.
- For example if you take a decision to purchase a particular material, the next question is how much.
- The next question is at what price.
- The next question is from whom... Like that there is no end.

Why Must Decisions Be Made?

- The decision theory helps to take a certain decision to have most satisfactory way of satisfying their needs.
- Decisions are made to achieve these goals and objectives.

Types of Decision Making Environment

- In general, decisions are classified as
- **Strategic decision**, which is related to the organization's outside environment,
- **administrative decisions** dealing with structuring resources and
- **operational decisions** dealing with day-to-day problems.

- Depending on the nature of the problem there are
 - a. **Programmed decisions**, to solve repetitive and well-structured problems, and
 - b. **Non-programmed decisions**, designed to solve non-routine, novel, illstructured problems.
- Depending on the scope, complexity and the number of people employed decision can be divided as
 - a. **Individual and**
 - b. **managerial** decisions.
- Depending on the sphere of interest, as **political**, **economic**, or **scientific** etc. decision can be divided as
 - a. **static** decision requiring only one decision for the planning horizon and
 - b. **dynamic decision** requiring a series of decisions for the planning horizon.

Steps In Decision Theory Approach

1. List the **viable alternatives (strategies)** that can be considered in the decision.
2. List all **future events** that can occur. These future events (not in the control of decision maker) are called as **states of nature**.
3. Construct a payoff table for each possible combination of alternative course of action and state of nature.
4. Choose the criterion that results in the largest payoff.

Decision Making Situations(Conditions)

Decision Making Under Certainty (DMUC)

- Decision making under certainty assumes that all relevant information required to make decision is certain in nature and is well known.
- It uses a deterministic model, with complete knowledge, stability and no ambiguity.
- To make decision, the manager will have to be quite aware of the strategies available and their payoffs and each strategy will have unique payoff resulting in certainty.
- The decision-making may be of single objective or of multiple objectives.

Decision making under Uncertainty

- In decision making under uncertainty no probability to each strategy is attached.
- In decision making under uncertainty, remember that no probabilities are attached to set of the states of nature.
- Sometimes we may have only positive elements in the given matrix, indicating that the company under any circumstances will have profit only.
- Sometimes, we may have negative elements, indicating potential loss.
- While solving the problem of decision making under uncertainty, we have two approaches, the first one is **pessimistic approach** and the second one is **optimistic approach**. Let us examine this by solving a problem.

Two criteria may be considered here.

- **Criterion of Optimism** and
- **Criterion of Pessimism.**

Criterion of Optimism

- The management of *XYZ* company is considering the use of a newly discovered chemical which, when added to detergents, will make the washing set, thus eliminating the necessity of adding softeners.
- The management is considering at present time, these three alternative strategies.
- S_1 = Add the new chemical to the currently marketed detergent DETER and sell it under label 'NEW IMPROVED DETER'.
- S_2 = Introduce a brand new detergent under the name of 'SUPER SOFT'
- S_3 = Develop a new product and enter the softener market under the name 'EXTRA WASH'.

- The management has decided for the time being that only one of the three strategies is economically feasible (under given market condition).
- The marketing research department is requested to develop a conditional payoff matrix for this problem.
- After conducting sufficient research, based on personal interviews and anticipating the possible reaction of the competitors, the marketing research department submits the payoff matrix given below. Select the optimal strategy.

State of nature.

Strategies.	N1	N2	N3
S1	15	12	18
S2	9	14	10
S3	13	4	26

- Here we determine the best possible outcome in each strategy, and then identify the **best of the best** outcome in order to select the optimal strategy.
- In the table given below the best of the best is written in the left hand side margin.

	State of nature.			<i>Best or Maximum outcome((Row maximum))</i>
Strategies.	N1	N2	N3	
S1	15	12	18	18

Criterion of Pessimism

- When criterion of pessimism is applied to solve the problem under uncertainty, first determine *worst possible outcome in each strategy* (row minimums), and select **the best of the worst outcome in** order to select the optimal strategy.
- The worst outcomes are shown in the left hand side margin.

	State of nature.			<i>Worst or minimum outcome (Row minimums)</i>
Strategies.	N1	N2	N3	
S1	15	12	18	12 MaxMin
S2	9	14	10	9
S3	13	4	26	4

Decision Making Under Risk (DMUR)

- Decision-making under risk (DMUR) describes a situation in which each strategy results in more than one outcomes or payoffs and the manager attaches a probability measure to these payoffs.
- This model covers the case when the manager projects two or more outcomes for each strategy and he or she knows, or is willing to assume, the relevant probability distribution of the outcomes.
- The following assumptions are to be made:
 - (1) Availability of more than one strategies,
 - (2) The existence of more than one states of nature,
 - (3) The relevant outcomes and
 - (4) The probability distribution of outcomes associated with each strategy.
- The optimal strategy in decision making under risk is identified by the strategy with highest expected utility (or highest expected value).

Example

A marketing manager of an insurance company has kept complete records of the sales effort of the sales personnel. These records contain data regarding the number of insurance policies sold and net revenues received by the company as a function of four different sales strategies. The manager has constructed the conditional payoff matrix given below, based on his records. (The state of nature refers to the number of policies sold). The number within the table represents utilities. Suppose you are a new salesperson and that you have access to the original records as well as the payoff matrix. Which strategy would you follow?

Solution

- As the decision is to be made under risk, multiplying the probability and utility and summing them up give the expected utility for the strategy.

<i>State of Nature</i>	N1	N2	N3	<i>Expected utility or expected payoffs</i>
Probability	0.2	0.5	0.3	
Strategy	Utility	Utility	Utility	
S1	4	6	10	$0.2 \times 4 + 0.5 \times 6 + 0.3 \times 10 = 6.8$
S2	6	5	9	$0.2 \times 6 + 0.5 \times 5 + 0.3 \times 9 = 6.4$
S3	2	10	8	$0.2 \times 2 + 0.5 \times 10 + 0.3 \times 8 = 7.8$

Criterion of Regret

- In this case, we have to determine the **regret matrix or opportunity loss matrix**.
- To find the opportunity loss matrix (column opportunity loss matrix), subtract all the elements of a column from the highest element of that column.
- The obtained matrix is known as *regret matrix*.
- While selecting the best strategy, we have to select such a strategy, whose opportunity loss is zero, *i.e.* zero regret.
- If we select any other strategy, then the regret is the element at that strategy.

- Rule for getting the regret matrix:
- In each column, identify the highest element and then subtract all the individual elements of that column, cell by cell, from the highest element to obtain the corresponding column of the regret matrix.
- To select the optimal strategy we first determine the maximum regret that the decision maker can experience for each strategy and then identify the maximum of the maximum regret values.

State of Nature				
Strategies	N1	N2	N3	Maximum regret.
Regret or Opportunity loss.				
S1	0	2	8	8 minimax. Regret
S2	6	0	16	16
S3	2	10	0	10

- Suppose two strategies have same minimax element, then the manager needs additional factors that influence his selection.
- Suppose two strategies have same minimax element, then the manager needs additional factors that influence his selection.

Chapter Five

Network Models

5.1. General network concepts

Introduction

- A project is a combination of various activities.
- In planning and scheduling the activities of large sized projects, the two network techniques — PERT and CPM — are used conveniently to estimate and evaluate the project completion time and control the resources to see that the project is completed within the stipulated time and at minimum possible cost.
- Initially, projects were represented by **milestone chart** and **bar chart**.
- But they had little use in controlling the project activities.
- **Bar chart** simply represents each activity by bars of length equal to the time taken on a common time scale.

- Networking algorithms

PERT AND CPM

- In PERT and CPM the milestones are represented as ***events***. Event or node is either starting of an activity or ending of an activity. Activity is represented by means of an arrow, which is resource consuming. Activity consumes resources like time, money and materials. Event will not consume any resource, but it simply represents either starting or ending of an activity. Event can also be represented by rectangles or triangles. When all activities and events in a project are connected logically and sequentially, they form a ***network***, which is the basic document in network-based management.

The basic steps for writing a network are:

(a) List out all the activities involved in a project. Say, for example, in building construction, the activities are:

(i) Site selection,

(ii) Arrangement of Finance,

(iii) Preparation of building plan,

(iv) Approval of plan by municipal authorities,

- (v) Purchase of materials,
- (vi) Digging of foundation,
- (vii) Filling up of foundation,
- (viii) Building superstructure,
- (ix) Fixing up of doorframes and window frames,

- (x) Roofing,
- (xi) Plastering,
- (xii) Flooring,
- (xiii) Electricity and water fittings,
- (xiv) Finishing.

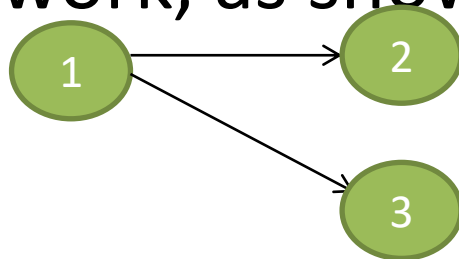
b) Once the activities are listed, they are arranged in sequential manner and in logical order. Forexample, foundation digging should come before foundation filling and so on.

(c) After arranging the activities in a logical sequence, their time is estimated and written against each activity. For example: Foundation digging: 10 days, or 1½ weeks.

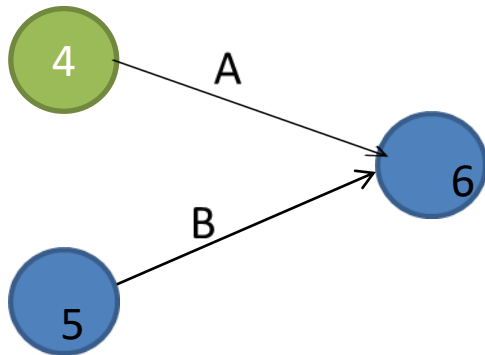
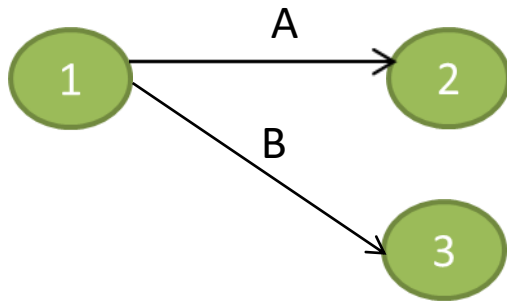
(d) Some of the activities do not have any logical relationship, in such cases; we can start those activities simultaneously.

- For example, foundation digging and purchase of materials do not have any logical relationship.
- Hence both of them can be started simultaneously. Suppose foundation digging takes 10 days and purchase of materials takes 7 days, both of them can be finished in 10 days.
- And the successive activity, say foundation filling, which has logical relationship with both of the above, can be started after 10 days.
- Otherwise, foundation digging and purchase of materials are done one after the other; filling of foundation should be started after 17 days.

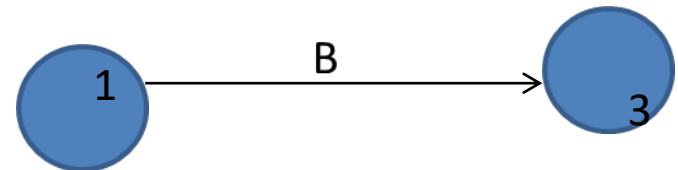
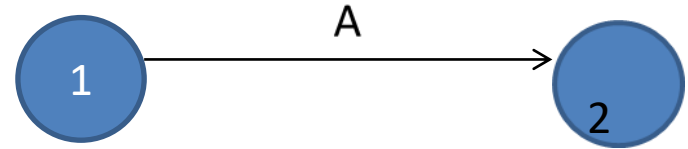
- Activities are added to the network, depending upon the logical relationship to complete the project network.
- Some of the points to be remembered while drawing the network are
(a) There must be only one beginning and one end for the network, as shown in figure below



Right



Wrong



- b) Event number should be written inside the circle or node (or triangle/square/rectangle etc). Activity name should be capital alphabetical letters and would be written above the arrow.
- c) While writing network, see that activities should not cross each other.
- d) While writing network, looping should be avoided. This is to say that the network arrows should move in one direction, i.e. starting from the beginning should move towards the end,
- e) When two activities start at the same event and end at the same event, they should be shown by means of a **dummy activity**
- Dummy activity is an activity, which simply shows the logical relationship and does not consume any resource. It should be represented by a dotted line.

- When the event is written at the tail end of an arrow, it is known as **tail event**.
- If event is written on the head side of the arrow it is known as **head event**.
- A tail event may have any number of arrows (activities) emerging from it.
- This is to say that an event may be a tail event to any number of activities.
- Similarly, a head event may be a head event for any number of activities.
- This is to say that many activities may conclude at one event.

Basic Difference Between PERT and CPM

i. PERT is event oriented and CPM is activity oriented.

- This is to say that while discussing about PERT network, we say that Activity 1-2, Activity 2-3 and so on.
- Or event 2 occurs after event 1 and event 5 occurs after event 3 and so on.
- While discussing CPM network, we say that Activity *A* follows activity *B* and activity *C* follows activity *B* and so on.
- Referring to the network shown in figure 9, we can discuss as under.
- PERT way: Event 1 is the predecessor to event 2 or event 2 is the successor to event 1.
- Events 3 and 4 are successors to event 2 or event 2 is the predecessor to events 3 and 4.

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- CPM way: Activity 1-2 is the predecessor to Activities 2-3 and 2-4

- ii. PERT activities are probabilistic in nature.
- The time required to complete the PERT activity cannot be specified correctly.
 - Because of uncertainties in carrying out the activity, the time cannot be specified correctly.
 - Say, for example, if you ask a contractor how much time it takes to construct the house, he may answer you that it may take 5 to 6 months.
 - This is because of his expectation of uncertainty in carrying out each one of the activities in the construction of the house.

- Another example is if somebody asks you how much time you require to reach railway station from your house, you may say that it may take 1 to 1½ hours.
- This is because you may think that you may not get a transport facility in time.
- Or on the way to station, you may come across certain work, which may cause delay in your journey from house to station. Hence PERT network is used when the activity times are probabilistic.

- There are three time estimates in PERT, they are:
 - (a) **OPTIMISTIC TIME:** Optimistic time is represented by **tO**. Here the estimator thinks that everything goes on well and he will not come across any sort of uncertainties and estimates lowest time as far as possible. He is optimistic in his thinking.
 - (b) **PESSIMISTIC TIME:** This is represented by **tP**. Here estimator thinks that everything goes wrong and expects all sorts of uncertainties and estimates highest possible time. He is pessimistic in his thinking.

- (c) **LIKELY TIME:** This is represented by t_L .
- This time is in between optimistic and pessimistic times. Here the estimator expects he may come across some sort of uncertainties and many a time the things will go right.
- So while estimating the time for a PERT activity, the estimator will give the three time estimates.

Based on the three time estimations, we can Compute

- Standard deviation = $(t_P - t_O)/6 = \sigma$, $t_P - t_O$ is known as range.
- Variance = $\{(t_P - t_O)/6\}^2 = \sigma^2$
- Expected Time or Average Time = $t_E = (t_O + 4t_L + t_P) / 6$
- These equations are very important in the calculation of PERT times.
- Hence the student has to remember these formulae.

Now let us see how to deal with the PERT problems.

- **Numbering of events:** Once the network is drawn the events are to be numbered.
- In PERT network, as the activities are given in terms of events, we may not experience difficulty.

Example

- A project consists of 9 activities and the three time estimates are given below. Find the project completion time (TE).
1. Write the network for the given project and find the project completion time?

Activities			Days		
	i	j	TO	TL	TP
	10	20	5	12	17
	10	30	8	10	13
	10	40	9	11	12
	20	30	5	8	9
	20	50	9	11	13
	40	60	14	18	22
	30	70	21	25	30
	60	70	8	13	17
	60	80	14	17	21
	70	80	6	9	12

Solution

- In PERT network, it is easy to write network diagram, because the successor and predecessor event relationships can easily be identified. While calculating the project completion time, we have to calculate t_e i.e. expected completion time for each activity from the given three-time estimates. In case, we calculate project completion time by using t_O or t_L or t_P separately, we will have three completion times. Hence it is advisable to calculate t_E expected completion time for each activity and then the project completion time. Now let us work out expected project completion time.

<i>Predecessor or event</i>	<i>Successor event</i>	Time in days			$TE = (tO + 4tL + tP)/6$	<i>Range $tP - tO$</i>	<i>S.D (σ) $(tP - tO) /6$</i>	<i>Variance σ_2</i>
		TO	TL	TP				
10	20	5	12	17	9.66 (10)	12	2	4
10	30	8	10	13	10.17 (10)	5	0.83	0.69
10	40	9	11	12	10.83 (11)	3	0.5	0.25
20	30	5	8	9	7.67 (8)	4	0.66	0.44
20	50	9	11	13	11.00 (11)	4	0.66	0.44
40	60	14	18	22	18.00 (18)	8	1.33	1.78
30	70	21	25	30	25.18 (25)	9	1.5	2.25